

第十一章 柱函数

本章基本要求

- 1. 掌握柱函数的基本理论;
- 2. 熟练掌握柱内、空心柱体、柱外区域上稳定场问题的求解方法;
- 3. 掌握柱坐标系中含时定解问题的求解方法;
- 4. 掌握球坐标系中含时定解问题的求解方法。

§11. 2 贝塞尔方程

1. 证明

- (1) 对于整数 n , 证明 $|J_n(z)| \cdot \frac{1}{n!} \left(\frac{|z|}{2}\right)^n e^{\left(\frac{|z|}{2}\right)^2}$, 且 $\lim_{n \rightarrow \infty} J_n(z) = 0$;
- (2) $\frac{d}{dz} \left[\frac{J_{-\nu}(z)}{J_\nu(z)} \right] = -\frac{2 \sin \nu \pi}{\pi z [J_\nu(z)]^2}$, 当且仅当 ν 为整数, $J_\nu(z)$ 和 $J_{-\nu}(z)$ 线性相关。

证: (1) 对于整数阶贝塞尔函数

$$J_n(z) = \sum_{k=0}^{\infty} (-1)^k \frac{1}{k!(n+k)!} \left(\frac{z}{2}\right)^{n+2k},$$

有

$$|J_n(z)| \cdot \left|\frac{z}{2}\right|^n \sum_{k=0}^{\infty} \frac{1}{k!(n+k)!} \left|\frac{z}{2}\right|^{2k} \cdot \frac{1}{n!} \left|\frac{z}{2}\right|^n \sum_{k=0}^{\infty} \frac{1}{k!(n+1)^k} \left|\frac{z}{2}\right|^{2k}$$

当 $n \neq 0$ 时,

$$|J_n(z)| \cdot \frac{1}{n!} \left|\frac{z}{2}\right|^n \exp\left(\frac{1}{4} \frac{|z|^2}{n+1}\right) \cdot \frac{1}{n!} \left|\frac{z}{2}\right|^n \exp\left(\left|\frac{z}{2}\right|^2\right)$$

证毕。

(2)

$$\frac{d}{dz} \left[\frac{J_{-\nu}(z)}{J_\nu(z)} \right] = \frac{J_\nu(z) J_{-\nu}'(z) - J_\nu'(z) J_{-\nu}(z)}{[J_\nu(z)]^2}, \tag{1}$$

又由 $J_{\pm \nu}(z)$ 所满足的微分方程

$$\frac{d}{dz} \left(z \frac{dJ_\nu}{dz} \right) + \left(z - \frac{\nu^2}{z} \right) J_\nu = 0, \tag{2}$$

$$\frac{d}{dz} \left(z \frac{dJ_{-\nu}}{dz} \right) + \left(z - \frac{\nu^2}{z} \right) J_{-\nu} = 0, \tag{3}$$

② $\times J_{-\nu}$ - ③ $\times J_\nu$ 得

$$J_{-\nu} \frac{d}{dz} \left(z \frac{dJ_\nu}{dz} \right) - J_\nu \frac{d}{dz} \left(z \frac{dJ_{-\nu}}{dz} \right) = 0,$$

即

$$z \left[J_{-\nu}(z) J_\nu'(z) - J_{-\nu}'(z) J_\nu(z) \right] = \text{常数}。 \tag{4}$$

将 $J_{\pm \nu}(z)$ 的级数表达式

$$J_{\pm \nu}(z) = \sum_{k=0}^{\infty} (-1)^k \frac{1}{k! \Gamma(\pm \nu + k + 1)} \left(\frac{z}{2}\right)^{\pm \nu + 2k}$$

代入④, 再令 $z \rightarrow 0$, 即得④式中的常数为 $2 \sin \nu \pi / \pi$, 即

$$\left[J_\nu(z) J_{-\nu}'(z) - J_\nu'(z) J_{-\nu}(z) \right] = -\frac{2 \sin \nu \pi}{\pi z} \tag{5}$$

⑤式表明, 当且仅当 ν 为整数时, J_ν 和 $J_{-\nu}$ 才是线性相关的。将⑤式代回①, 即得

$$\frac{d}{dz} \left[\frac{J_{-\nu}(z)}{J_\nu(z)} \right] = -\frac{2 \sin \nu \pi}{\pi z [J_\nu(z)]^2}。$$

2. 计算不定积分 $\int J_3(x)dx$ 。

解：利用递推公式 $\frac{d}{dx} \left[\frac{J_m(x)}{x^m} \right] = -\frac{J_{m+1}(x)}{x^m}$,

$$\begin{aligned}\int J_3(x)dx &= \int x^2 [x^{-2}J_3(x)]dx = -\int x^2 d[x^{-2}J_2(x)] \\ &= -J_2(x) + \int 2x^{-1}J_2(x)dx \\ &= -J_2(x) - 2x^{-1}J_1(x) + C\end{aligned}$$

再利用递推公式 $J_2(x) = (2/x)J_1(x) - J_0(x)$ ，有

$$\begin{aligned}\int J_3(x)dx &= -2x^{-1}J_1(x) + J_0(x) - 2x^{-1}J_1(x) + C \\ &= J_0(x) - 4x^{-1}J_1(x) + C\end{aligned}$$

3. 在区间[0,1]上,第一类齐次边界条件下,用零阶贝塞尔函数把 $f(x)=1$ 展开为傅里叶-贝塞尔级数。

解: 通常以 x 作为贝塞尔函数的宗量, 此处先将 $f(x)$ 中的 x 换为 ρ ,

$$f(x)=1 \Rightarrow f(\rho)=1$$

展开式为

$$f(\rho) = \sum_{n=1}^{\infty} f_n J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right), \quad \rho_0=1。$$

展开系数为

$$f_n = \frac{1}{[N_n^{(0)}]^2} \int_0^{\rho_0} f(\rho) J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \rho d\rho。$$

在第一类边界条件下

$$[N_n^{(0)}]^2 = \frac{1}{2} \cdot 1^2 \cdot [J_1(x_n^{(0)})]^2 = \frac{1}{2} [J_1(x_n^{(0)})]^2。$$

其中 $x_n^{(0)}$ 为 $J_0(x)$ 的第 n 个零点。从而

$$\begin{aligned} f_n &= \frac{2}{[J_1(x_n^{(0)})]^2} \int_0^1 J_0(x_n^{(0)} \rho) \rho d\rho \\ &= \frac{2}{[J_1(x_n^{(0)})]^2} \cdot \frac{1}{(x_n^{(0)})^2} \int_0^1 J_0(x_n^{(0)} \rho) (x_n^{(0)} \rho) d(x_n^{(0)} \rho) \\ &= \frac{2}{[x_n^{(0)} J_1(x_n^{(0)})]^2} \cdot (x_n^{(0)} \rho) J_1(x_n^{(0)} \rho) \Big|_0^1 = \frac{2}{x_n^{(0)} J_1(x_n^{(0)})} \end{aligned}$$

故

$$1 = \sum_{n=1}^{\infty} \frac{2}{x_n^{(0)} J_1(x_n^{(0)})} J_0(x_n^{(0)} \rho) \Rightarrow 1 = \sum_{n=1}^{\infty} \frac{2}{x_n^{(0)} J_1(x_n^{(0)})} J_0(x_n^{(0)} x)。$$

4. 半径为 ρ_0 、高为 L 的圆柱体,下底和侧面保持零度,上底温度分布为 $f(\rho) = \rho^2$,求柱体内各点的稳恒温度分布。

解: 采用柱坐标系,极点在下底中心, z 轴沿着圆柱的轴,定解问题表为

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = 0, \quad u|_{z=L} = f(\rho) = \rho^2. \end{cases} \quad (1)$$

本题是圆柱内部的拉普拉斯方程定解问题。因柱侧是齐次的第二类边界条件,且问题与 φ 角无关,因而,定解问题①经分离变数后,由零阶贝塞尔方程与径向边界条件构成本征问题

$$\begin{cases} x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + x^2 R = 0, & x = \sqrt{\mu} \rho \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty. \end{cases} \quad (2)$$

②的本征值是 $\sqrt{\mu_n^{(0,1)}} = \frac{x_n^{(0)}}{\rho_0}$, 本征函数是 $J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)$, $n=1,2,\dots$, 其中 $x_n^{(0)}$ 是零数阶贝塞尔函数的第 n 个零点。这时, Z 方向方程 $Z'' - \mu Z = 0$ 的解

$$Z(\rho) = A_n e^{x_n^{(0)} z / \rho_0} + B_n e^{-x_n^{(0)} z / \rho_0},$$

从而,定解问题①有限解的一般形式是

$$u(\rho, z) = \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} z / \rho_0} + B_n e^{-x_n^{(0)} z / \rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right). \quad (3)$$

为确定系数 A_n 和 B_n , 将③代入①中上下底面边界条件, 有

$$\begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = 0, \\ \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} L / \rho_0} + B_n e^{-x_n^{(0)} L / \rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = \rho^2, \end{cases}$$

由此, 得

$$\begin{cases} A_n + B_n = 0, \\ A_n e^{x_n^{(0)} L / \rho_0} + B_n e^{-x_n^{(0)} L / \rho_0} \end{cases} \quad (4)$$

$$= \frac{2}{\rho_0^2 [J_1(x_n^{(0)})]^2} \int_0^{\rho_0} \rho^2 J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \rho d\rho \equiv f_n. \quad (5)$$

由④有 $B_n = -A_n$ ，从而⑤式成为

$$2A_n \operatorname{sh}(x_n^{(0)} L / \rho_0) = f_n \quad \Rightarrow \quad A_n = \frac{f_n}{2 \operatorname{sh}(x_n^{(0)} L / \rho_0)}$$

⑤式中的积分

$$\begin{aligned} \int_0^{\rho_0} \rho^3 J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) d\rho &= \left(\frac{\rho_0}{x_n^{(0)}}\right)^4 \int_0^{\rho_0} \left(\frac{x_n^{(0)}}{\rho_0} \rho\right)^3 J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) d\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \\ &= \left(\frac{\rho_0}{x_n^{(0)}}\right)^4 \int_0^{x_0} x^3 J_0(x) dx \\ &= \left(\frac{\rho_0}{x_n^{(0)}}\right)^4 \left[(x_n^{(0)})^3 J_1(x_n^{(0)}) - 4x_n^{(0)} J_1(x_n^{(0)}) \right] \end{aligned}$$

其中积分 $\int_0^{x_0} x^3 J_0(x) dx$ 的计算利用了教材本节例 1 的结果。这样

$$f_n = \frac{2\rho_0^2}{x_n^{(0)} J_1(x_n^{(0)})} \left(1 - \frac{4}{(x_n^{(0)})^2}\right), \quad A_n = \frac{\rho_0^2}{x_n^{(0)} J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L / \rho_0)} \left[1 - \frac{4}{(x_n^{(0)})^2}\right].$$

从而

$$u(\rho, z) = 2\rho_0^2 \sum_{n=1}^{\infty} \frac{1 - 4/(x_n^{(0)})^2}{x_n^{(0)} J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L / \rho_0)} \operatorname{sh}\left(\frac{x_n^{(0)}}{\rho_0} z\right) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right).$$

5. 半径为 ρ_0 而高为 L 的圆柱体, 下底温度分布为 $u_0\rho^2$, 上底温度保持为 u_1 , 侧面绝热, 求柱体内的稳恒温度分布.

解: 根据题意, 定解问题是

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = u_0\rho^2, \quad u|_{z=L} = u_1. \end{cases} \quad (1)$$

类似上题, 本征值问题仍由零阶贝塞尔方程与径向边界条件构成

$$\begin{cases} x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + x^2 R = 0, & x = \sqrt{\mu} \rho \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty, \end{cases}$$

本征值是 $\sqrt{\mu_n^{(0,2)}} = \frac{x_n^{(0,2)}}{\rho_0} = \frac{x_n^{(1)}}{\rho_0}$, 本征函数是 $J_0\left(\frac{x_n^{(1)}}{\rho_0}\rho\right)$, $n=1, 2, \dots$ 。其中 $x_n^{(1)}$ 是 1 阶贝

塞尔函数的第 n 个零点。由于 $J_1(x)$ 的第 0 个零点是 $x=0$, 从而第 0 个本征值 $\mu_0^{(0,2)} = 0$ 。这时,

Z 方向方程 $\begin{cases} Z'' = 0, & \mu = 0 \\ Z'' - \mu Z = 0, & \mu \neq 0 \end{cases}$ 的解是

$$\begin{cases} Z(\rho) = A_0 + B_0 z, & \mu = 0 \\ Z(\rho) = A_n e^{x_n^{(1)} z / \rho_0} + B_n e^{-x_n^{(1)} z / \rho_0}, & \mu \neq 0 \end{cases}$$

从而, 定解问题①有限解的一般形式是

$$u(\rho, z) = A_0 + B_0 z + \sum_{n=1}^{\infty} (A_n e^{x_n^{(1)} z / \rho_0} + B_n e^{-x_n^{(1)} z / \rho_0}) J_0\left(\frac{x_n^{(1)}}{\rho_0}\rho\right). \quad (2)$$

为确定系数 A_n 和 B_n , 将②代入①中上下底面边界条件

$$\begin{cases} A_0 + \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(1)}}{\rho_0}\rho\right) = u_0\rho^2, \\ A_0 + B_0 L + \sum_{n=1}^{\infty} (A_n e^{x_n^{(1)} L / \rho_0} + B_n e^{-x_n^{(1)} L / \rho_0}) J_0\left(\frac{x_n^{(1)}}{\rho_0}\rho\right) = u_1. \end{cases} \quad (3)$$

将③式右边按 $J_0\left(\frac{x_n^{(1)}}{\rho_0}\rho\right)$ 展为傅里叶-贝塞尔级数, 比较两边系数, 得

$$\begin{cases} A_0 = \frac{2}{\rho_0^2} \int_0^{\rho_0} u_0 \rho^2 \cdot \rho d\rho = \frac{1}{2} u_0 \rho_0^2, \\ A_0 + B_0 L = \frac{2}{\rho_0^2} \int_0^{\rho_0} u_1 \cdot \rho d\rho = u_1. \end{cases} \quad (4)$$

$$\begin{cases} A_n + B_n = \frac{2}{\rho_0^2 [J_0(x_n^{(1)})]^2} \int_0^{\rho_0} u_0 \rho^2 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho = \frac{4u_0 \rho_0^2}{(x_n^{(1)})^2 J_0(x_n^{(1)})}, \\ A_n e^{x_n^{(1)} L / \rho_0} + B_n e^{-x_n^{(1)} L / \rho_0} \\ = \frac{2}{\rho_0^2 [J_0(x_n^{(1)})]^2} \int_0^{\rho_0} u_1 J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right) \rho d\rho = 0. \end{cases} \quad (5)$$

解④⑤得

$$\begin{cases} A_0 = \frac{1}{2} u_0 \rho_0^2, \\ B_0 = (u_1 - \frac{1}{2} u_0 \rho_0^2) / L. \end{cases} \quad \begin{cases} A_n = -\frac{2u_0 \rho_0^2 e^{-x_n^{(1)} L / \rho_0}}{(x_n^{(1)})^2 J_0(x_n^{(1)}) \text{sh}(x_n^{(1)} L / \rho_0)}, \\ B_n = \frac{2u_0 \rho_0^2 e^{x_n^{(1)} L / \rho_0}}{(x_n^{(1)})^2 J_0(x_n^{(1)}) \text{sh}(x_n^{(1)} L / \rho_0)}. \end{cases}$$

从而, 解是

$$\begin{aligned} u(\rho, z) = & \frac{u_0}{2} \rho_0^2 + \frac{u_1 - (u_0 \rho_0^2 / 2)}{L} z + \sum_{n=1}^{\infty} \frac{4u_0 \rho_0^2}{(x_n^{(1)})^2 J_0(x_n^{(1)}) \text{sh}(x_n^{(1)} L / \rho_0)} \\ & \times \text{sh}[x_n^{(1)} (L - z) / \rho_0] J_0\left(\frac{x_n^{(1)}}{\rho_0} \rho\right). \end{aligned}$$

6.圆柱体半径为 ρ_0 而高为 L , 上底有均匀分布的强度为 q_0 的热流进入, 下底则有同样的热流流出, 柱侧保持温度零度, 求柱体内的稳恒温度分布。

解: 根据题意, 定解问题是

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ \left. \frac{\partial u}{\partial z} \right|_{z=0} = \frac{q_0}{k}, \quad \left. \frac{\partial u}{\partial z} \right|_{z=L} = \frac{q_0}{k}. \end{cases} \quad (1)$$

定解问题①经分离变数后, 由零阶贝塞尔方程与径向边界条件构成本征问题

$$\begin{cases} x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + x^2 R = 0, \quad x = \sqrt{\mu} \rho \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty, \end{cases} \quad (2)$$

②的本征值是 $\sqrt{\mu_n^{(0,1)}} = \frac{x_n^{(0)}}{\rho_0}$, 本征函数是 $J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right)$, $n=1, 2, \dots$ 。其中 $x_n^{(0)}$ 是零数阶贝塞尔函数的第 n 个零点。这时, Z 方向方程 $Z'' - \mu Z = 0$ 的解是

$$Z(\rho) = A_n e^{x_n^{(0)} z / \rho_0} + B_n e^{-x_n^{(0)} z / \rho_0},$$

从而, 定解问题①有限解的一般形式是

$$u(\rho, z) = \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} z / \rho_0} + B_n e^{-x_n^{(0)} z / \rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right). \quad (3)$$

为确定系数 A_n 和 B_n , 将③代入①中上下底面边界条件, 得到

$$\begin{cases} \sum_{n=1}^{\infty} (A_n - B_n) \left(\frac{x_n^{(0)}}{\rho_0} \right) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = \frac{q_0}{k}, \\ \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} L / \rho_0} - B_n e^{-x_n^{(0)} L / \rho_0}) \left(\frac{x_n^{(0)}}{\rho_0} \right) J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) = \frac{q_0}{k}. \end{cases} \quad (4)$$

由④, 我们有

$$\begin{cases} A_n - B_n = -\frac{2q_0}{k x_n^{(0)} \rho_0 [J_1(x_n^{(0)})]^2} \int_0^{\rho_0} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \rho d\rho = \frac{2q_0 \rho_0}{k (x_n^{(0)})^2 J_1(x_n^{(0)})}, \\ A_n e^{x_n^{(0)} L / \rho_0} - B_n e^{-x_n^{(0)} L / \rho_0} = \frac{2q_0 \rho_0}{k (x_n^{(0)})^2 J_1(x_n^{(0)})}. \end{cases} \quad (5)$$

由⑤解出

$$\begin{cases} A_n = \frac{q_0 \rho_0}{k (x_n^{(0)})^2 J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L / \rho_0)} (1 - e^{-x_n^{(0)} L / \rho_0}), \\ B_n = \frac{q_0 \rho_0}{k (x_n^{(0)})^2 J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L / \rho_0)} (1 - e^{x_n^{(0)} L / \rho_0}). \end{cases}$$

从而, 解是

$$u(\rho, z) = \sum_{n=1}^{\infty} \frac{2q_0 \rho_0}{k (x_n^{(0)})^2 J_1(x_n^{(0)}) \operatorname{sh}(x_n^{(0)} L / \rho_0)} \left[\operatorname{ch} \frac{x_n^{(0)}}{\rho_0} z - \operatorname{ch} \frac{x_n^{(0)}}{\rho_0} (L - z) \right] J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right).$$

7. 半径为 ρ_0 的圆形膜,边缘固定,初始形状是旋转抛物面 $u|_{t=0}=u_0(1-\rho^2/\rho_0^2)$,初始速度为零,求解膜的振动情况.

解: 定解问题是

$$\begin{cases} u_{tt} - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho}) = 0, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ u|_{t=0} = u_0(1 - \frac{\rho^2}{\rho_0^2}), \quad u_t|_{t=0} = 0. \end{cases} \quad (1)$$

令: $u(\rho, t) = T(t)R(\rho)$, 代入定解问题①, 分离变数, 得

$$T'' + k^2 a^2 T = 0, \quad (2)$$

及本征值问题

$$\begin{cases} R'' + \frac{1}{\rho}R' + k^2 R = 0 \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty, \end{cases} \Rightarrow \begin{cases} \rho^2 R'' + \rho R' + k^2 \rho^2 R = 0 \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty, \end{cases} \quad (3)$$

③的本征函数是 $R_n(\rho) = J_0(k_n^{(0)}\rho)$, 本征值是 $k_n^{(0)} = \frac{x_n^{(0)}}{\rho_0}$, $n=1, 2, \dots$ 。其中 $x_n^{(0)}$ 是零数阶贝塞尔函数的第 n 个零点。方程②的解是 $T(t) = A_n \cos k_n^{(0)}at + B_n \sin k_n^{(0)}at$, 于是, 问题①的一般解是

$$u(\rho, t) = \sum_{n=1}^{\infty} (A_n \cos k_n^{(0)}at + B_n \sin k_n^{(0)}at) J_0(k_n^{(0)}\rho) \quad (4)$$

将④代入①中初始条件以确定系数, 由 $u_t|_{t=0} = 0$, 有 $B_n = 0$ 。

由 $u|_{t=0} = \sum_{n=1}^{\infty} A_n J_0(k_n^{(0)}\rho) = u_0(1 - \frac{\rho^2}{\rho_0^2})$, 有

$$A_n = \frac{2u_0}{\rho_0^2 [J_1(x_n^{(0)})]^2} \int_0^{\rho_0} \left(1 - \frac{\rho^2}{\rho_0^2}\right) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \rho d\rho,$$

其中 $\int_0^{\rho_0} J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \rho d\rho = \frac{\rho_0^2}{(x_n^{(0)})^2} \int_0^{x_n^{(0)}} x J_0(x) dx = \frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}),$

$$\begin{aligned} \int_0^{\rho_0} \rho^2 J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \rho d\rho &= \frac{\rho_0^4}{(x_n^{(0)})^4} \int_0^{x_n^{(0)}} x^3 J_0(x) dx \\ &= \frac{\rho_0^4}{(x_n^{(0)})^4} \left[x^3 J_1(x) - 4x J_1(x) + 2x^2 J_0(x) \right]_0^{x_n^{(0)}} \\ &= \frac{\rho_0^4}{x_n^{(0)}} J_1(x_n^{(0)}) - \frac{4\rho_0^4}{(x_n^{(0)})^3} J_1(x_n^{(0)})。 \end{aligned}$$

从而

$$\begin{aligned} A_n &= \frac{2u_0}{\rho_0^2 [J_1(x_n^{(0)})]^2} \left[\frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}) - \frac{\rho_0^2}{x_n^{(0)}} J_1(x_n^{(0)}) + \frac{4\rho_0^2}{(x_n^{(0)})^3} J_1(x_n^{(0)}) \right] \\ &= \frac{8u_0}{(x_n^{(0)})^3 J_1(x_n^{(0)})} \end{aligned}$$

结果

$$u(\rho, t) = 8u_0 \sum_{n=1}^{\infty} \frac{1}{(x_n^{(0)})^3 J_1(x_n^{(0)})} J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \cos \frac{x_n^{(0)}at}{\rho_0}。$$

8. 半径为 ρ_0 的半圆形膜,边缘固定,求其本征频率和本征振动。

解: 采用极坐标, 定解问题是

$$\begin{cases} u_{tt} - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\varphi\varphi}) = 0, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ u|_{\varphi=0} = u|_{\varphi=\pi} = 0. \quad (0 \leq \rho \leq \rho_0, 0 \leq \varphi \leq \pi) \end{cases} \quad (1)$$

令 $u(\rho, t) = R(\rho)\Phi(\varphi)T(t)$, 代入定解问题①, 分离变数, 得

$$T'' + k^2 a^2 T = 0, \quad (2)$$

及本征值问题

$$(1) \begin{cases} \Phi'' + \lambda \Phi = 0 \\ \Phi|_{\varphi=0} = \Phi|_{\varphi=\pi} = 0. \end{cases} \quad (3)$$

本征函数是 $\Phi(\varphi) = D \sin m\varphi$. 本征值是 $\lambda = m^2$, ($m = 0, 1, 2, \dots$)

$$(2) \begin{cases} \rho^2 \frac{d^2 R}{d\rho^2} + \rho \frac{dR}{d\rho} + (k^2 \rho^2 - m^2) R = 0 \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty \end{cases} \quad (4)$$

本征函数是 $R_n(\rho) = J_0(k_n^{(m)} \rho)$, 本征值是 $k_n^{(m)} = \frac{x_n^{(m)}}{\rho_0}$, $n=1, 2, \dots$ 。其中 $x_n^{(m)}$ 是 m 数阶贝塞尔函数的第 n 个零点。方程②的解是

$$T(t) = A_n^{(m)} \cos \frac{x_n^{(m)}}{\rho_0} at + B_n^{(m)} \sin \frac{x_n^{(m)}}{\rho_0} at。$$

本征圆频率是 $\omega_n^{(m)} = \frac{x_n^{(m)} a}{\rho_0}$, 本征振动是

$$u_n^{(m)}(\rho, \varphi, t) = J_m\left(\frac{x_n^{(m)}}{\rho_0} \rho\right) \sin m\varphi \left[A_n^{(m)} \cos \frac{x_n^{(m)}}{\rho_0} at + B_n^{(m)} \sin \frac{x_n^{(m)}}{\rho_0} at \right]。$$

9. 求长圆柱形轴块的临界半径。(“临界”一词的意义见 § 8.1 习题 8.)

解: 取柱坐标系, 记中子的增殖速率为 βu , 铀块柱体侧面加以防护, 无中子向外泄露, 又为长圆柱, 且问题与 φ 无关, 这样, 定解问题是

$$\begin{cases} u_t - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho}) = \beta u, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty. \end{cases} \quad (1)$$

令 $u(\rho, t) = R(\rho)T(t)$, 代入定解问题①, 分离变数, 得

$$T' + k^2 a^2 T = 0, \quad (2)$$

及本征问题

$$\begin{cases} \rho^2 R'' + \rho R' + \frac{(\beta + k^2 a^2) \rho^2}{a^2} R = 0 \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} = \text{有限值}, \end{cases} \quad (3)$$

③的本征函数是 $R_n(\rho) = BJ_0(\frac{\sqrt{\beta + (k_n^{(0)})^2 a^2}}{a} \rho)$, 本征值 $k_n^{(0)}$ 由

$$\begin{aligned} \frac{\sqrt{\beta + (k_n^{(0)})^2 a^2}}{a} \rho_0 &= x_n^{(0)}, \\ - (k_n^{(0)})^2 a^2 &= \beta - \frac{(x_n^{(0)})^2 a^2}{\rho_0^2} \quad (n=1, 2, \dots) \end{aligned}$$

确定。其中 $x_n^{(0)}$ 是零数阶贝塞尔函数的第 n 个零点。方程②的解是 $T(t) = A_n^{(0)} e^{-(k_n^{(0)})^2 a^2 t}$, 由此

可知, 只要有一个 $x_n^{(0)}$ 满足 $\beta > \frac{(x_n^{(0)})^2 a^2}{\rho_0^2}$, 则中子浓度将随时间指数指数增长, 铀块将爆炸。

而满足 $\beta - \frac{(x_n^{(0)})^2 a^2}{\rho_0^2} = 0$, 即 $\rho_0 = \frac{x_n^{(0)} a}{\sqrt{\beta}}$ 称为临界半径。 $x_n^{(0)}$ 中最小的是 $x_1^{(0)} = 2.405$, 从而

长圆柱铀块的临界半径是 $2.405 \frac{a}{\sqrt{\beta}}$ 。

10. 把温度为室温 u_0 的样品放入温度为 U_0 的烘炉内保温,但是,样品里的温度不可能立刻就转变为 U_0 ,它与 U_0 之差随着时间而衰减,今约定把这差值降到初始值的 $1/e$ 之时作为保温开始时间,试计算圆柱形样品放入烘炉多长时间才可计算保温时间。

解: 取柱坐标, 设柱体半径为 ρ_0 , 高为 L , 问题与 φ 无关, 则定解问题为

$$\begin{cases} u_t - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + u_{zz}) = 0, \\ u|_{\rho=\rho_0} = U_0, \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = U_0, \quad u|_{z=L} = U_0, \\ u|_{t=0} = u_0. \end{cases} \quad (1)$$

将边值齐次化, 令 $u = v + U_0$, ①转化为 v 的定解问题

$$\begin{cases} v_t - a^2(v_{\rho\rho} + \frac{1}{\rho}v_{\rho} + v_{zz}) = 0, \\ v|_{\rho=\rho_0} = 0, \quad v|_{\rho=0} < \infty, \\ v|_{z=0} = 0, \quad v|_{z=L} = 0, \\ v|_{t=0} = u_0 - U_0. \end{cases} \quad (2)$$

②之解的形式为

$$J_0(\sqrt{k^2 - h^2}\rho) \sin hz \cdot e^{-k^2 a^2 t}.$$

这里, $h = \frac{m\pi}{L}$, $\sqrt{k^2 - h^2}\rho_0 = x_n^{(0)}$ 。于是, ②的一般解为

$$v(\rho, z, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{np} J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \sin \frac{m\pi z}{L} \exp\left\{-\left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{m^2\pi^2}{L^2}\right]a^2 t\right\}.$$

其中 $n=1$ 和 $m=1$ 的一项随时间减小最慢, 可将此项降到初始值的 $1/e$ 作为保温开始时间, 即

$$\exp\left\{-\left[\left(\frac{x_n^{(0)}}{\rho_0}\right)^2 + \frac{m^2\pi^2}{L^2}\right]a^2 t\right\} = \frac{1}{e}.$$

由此解得

$$t = \frac{1}{a^2 \left[(2.405/\rho_0)^2 + (\pi/L)^2 \right]}.$$

11. 匀质圆柱, 半径 ρ_0 、高 L , 上下底面固定, 侧面自由, 初始位移为零, 初始速度为 $u_0 \rho^2 z$, 求柱内各处的振动情况。

解: 定解问题

$$\begin{cases} u_{tt} - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + u_{zz}) = 0, \\ u_{\rho}|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = 0, \quad u|_{z=L} = 0 \\ u|_{t=0} = 0, \quad u_t|_{t=0} = u_0 \rho^2 z. \end{cases} \quad (1)$$

柱内有限解是

$$u(\rho, z, t) = \sum_{n=0}^{\infty} \sum_{p=1}^{\infty} \left(A \cos k_n^{(0)} at + B \sin k_n^{(0)} at \right) J_0 \left(\sqrt{(k_n^{(0)})^2 - \frac{p^2 \pi^2}{L^2}} \rho \right) \sin \frac{p\pi z}{L}, \quad (2)$$

其中 $k_n^{(0)}$ 由 $\sqrt{(k_n^{(0)})^2 - \frac{p^2 \pi^2}{L^2}} \rho_0 = x_n^{(1)}$, 即由 $k_n^{(0)} = \sqrt{\frac{(x_n^{(1)})^2}{\rho_0^2} + \frac{p^2 \pi^2}{L^2}}$ 确定。将②代入①中初值以确定系数

$$u|_{t=0} = \sum_{n=0}^{\infty} \sum_{p=1}^{\infty} A J_0 \left(\sqrt{(k_n^{(0)})^2 - \frac{p^2 \pi^2}{L^2}} \rho \right) \sin \frac{p\pi z}{L} = 0 \quad (3)$$

$$u_t|_{t=0} = \sum_{n=0}^{\infty} \sum_{p=1}^{\infty} (k_n^{(0)} a B) J_0 \left(\sqrt{(k_n^{(0)})^2 - \frac{p^2 \pi^2}{L^2}} \rho \right) \sin \frac{p\pi z}{L} = u_0 \rho^2 z \quad (4)$$

由③ $A = 0$ 。由④

$$k_n^{(0)} a B = \frac{1}{(N_n^{(0)})^2} \int_0^{\rho_0} \rho^2 J_0 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right) \rho d\rho \cdot \frac{2}{L} \int_0^L z \sin \frac{p\pi z}{L} dz。$$

其中, $(N_n^{(0)})^2 = \frac{1}{2} \rho_0^2 [J_0(x_n^{(1)})]^2$ 以及

$$\begin{aligned} & \int_0^{\rho_0} \rho^2 J_0 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right) \rho d\rho \\ &= \left(\frac{\rho_0}{x_n^{(1)}} \right)^4 \left[\left(\frac{x_n^{(1)}}{\rho_0} \rho \right)^3 J_1 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right) - 4 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right) J_1 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right) + 2 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right)^2 J_0 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right) \right]_0^{\rho_0} \\ &= \frac{2\rho_0^4}{(x_n^{(1)})^2} J_0(x_n^{(1)}), \end{aligned}$$

$$\frac{2}{L} \int_0^L z \sin \frac{p\pi z}{L} dz = (-1)^{p+1} \frac{2L}{p\pi}。$$

这样

$$B = \frac{(-1)^{p+1} 8u_0 L \rho_0^2}{ap\pi (x_n^{(1)})^2 J_0(x_n^{(1)})} \cdot \frac{1}{\sqrt{(x_n^{(1)}/\rho_0)^2 + (p\pi/L)^2}}$$

从而, 解是

$$\begin{aligned} u(\rho, z, t) &= \sum_{n=0}^{\infty} \sum_{p=1}^{\infty} \frac{(-1)^{p+1} 8u_0 L \rho_0^2}{ap\pi (x_n^{(1)})^2 J_0(x_n^{(1)})} \cdot \frac{1}{\sqrt{(x_n^{(1)}/\rho_0)^2 + (p\pi/L)^2}} \\ &\quad \times \sin \sqrt{(x_n^{(1)}/\rho_0)^2 + (p\pi/L)^2} at \cdot \sin \frac{p\pi}{L} z \cdot J_0 \left(\frac{x_n^{(1)}}{\rho_0} \rho \right)。 \end{aligned}$$

12. 半径为 ρ_0 的圆形膜, 边缘固定, 每单位质量上的作用力为 $f = A \sin \omega t$. 求解膜的振动情况。

解: 定解问题是

$$\begin{cases} u_{tt} - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho}) = \frac{A \sin \omega t}{p}, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty. \end{cases} \quad (1)$$

其中 p 是膜每单位面积的质量. 这是无初值问题, 可设解为 $u(\rho, t) = v(\rho) \sin \omega t$. 将其代入定解问题①, 有

$$\begin{cases} -\omega^2 v - a^2(v_{\rho\rho} + \frac{1}{\rho}v_{\rho}) = \frac{A}{p}, \\ v|_{\rho=\rho_0} = 0, \quad v|_{\rho=0} < \infty, \end{cases} \Rightarrow \begin{cases} v_{\rho\rho} + \frac{1}{\rho}v_{\rho} + \frac{\omega^2}{a^2}v = -\frac{A}{pa^2}, \\ v|_{\rho=\rho_0} = 0, \quad v|_{\rho=0} < \infty, \end{cases} \quad (2)$$

令 $v(\rho) = w(\rho) - \frac{A}{pa^2}$, 代入定解问题②, ②转换成 w 的定解问题

$$\begin{cases} w_{\rho\rho} + \frac{1}{\rho}w_{\rho} + \frac{\omega^2}{a^2}w = 0, \\ w|_{\rho=\rho_0} = \frac{A}{p\omega^2}, \quad w|_{\rho=0} < \infty, \end{cases} \quad (3)$$

③的一般解是 $w(\rho) = BJ_0\left(\frac{\omega}{a}\rho\right)$, 代入③中边值, 有

$$BJ_0\left(\frac{\omega}{a}\rho_0\right) = \frac{A}{p\omega^2}, \Rightarrow B = \frac{A}{p\omega^2} \frac{1}{J_0\left(\frac{\omega}{a}\rho_0\right)},$$

从而

$$w(\rho) = \frac{A}{p\omega^2} \frac{1}{J_0\left(\frac{\omega}{a}\rho_0\right)} J_0\left(\frac{\omega}{a}\rho\right).$$

这样, 问题的解是

$$u(\rho, t) = \frac{A}{p\omega^2} \left[\frac{J_0\left(\frac{\omega}{a}\rho\right)}{J_0\left(\frac{\omega}{a}\rho_0\right)} - 1 \right] \sin \omega t.$$

13. 匀质圆柱, 半径 ρ_0 、高 L , 试求解圆柱内部的热传导问题.

$$\begin{cases} u_t - a^2 \Delta u = 0, \\ u|_{\rho=\rho_0} = u_1 z^2, \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = 0, \quad \frac{\partial u}{\partial z}|_{z=L} = 2Lu_1, \\ u|_{t=0} = u_0. \end{cases} \quad (1)$$

其中 u_0 、 u_1 为常数。

解: 令 $u(\rho, z, t) = v(\rho, z, t) + u_1 z^2$, 代入定解问题①, ①转换成 v 的定解问题

$$\begin{cases} v_t - a^2 \Delta v = 2u_1, \\ v|_{\rho=\rho_0} = 0, \quad v|_{\rho=0} < \infty, \\ v|_{z=0} = 0, \quad \frac{\partial v}{\partial z}|_{z=L} = 0, \\ v|_{t=0} = u_0 - u_1 z^2. \end{cases} \quad (2)$$

令 $v(\rho, z, t) = X(\rho, z, t) + Y(\rho, z, t)$, 代入定解问题②, ②分解为 X 、 Y 两个定解问题

$$\begin{cases} X_t - a^2 \Delta X = 0, \\ X|_{\rho=\rho_0} = 0, \quad X|_{\rho=0} < \infty, \\ X|_{z=0} = 0, \quad \frac{\partial X}{\partial z}|_{z=L} = 0, \\ X|_{t=0} = u_0 - u_1 z^2. \end{cases} \quad (3)$$

$$\begin{cases} Y_t - a^2 \Delta Y = 2u_1, \\ Y|_{\rho=\rho_0} = 0, \quad Y|_{\rho=0} < \infty, \\ Y|_{z=0} = 0, \quad \frac{\partial Y}{\partial z}|_{z=L} = 0, \\ Y|_{t=0} = 0. \end{cases} \quad (4)$$

先解 X 的定解问题③, 其一般解是

$$\begin{aligned} X(\rho, z, t) = & \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} A_{np} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \sin \frac{(p + \frac{1}{2})\pi z}{L} \\ & \times \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t \right\} \end{aligned} \quad (5)$$

将⑤代入③中初值，以确定系数

$$X|_{t=0}=\sum_{n=1}^{\infty}\sum_{p=0}^{\infty}A_{np}J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)\sin\frac{(p+\frac{1}{2})\pi z}{L}=u_0-u_1z^2。$$

将右端按 $J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)\sin\frac{(p+\frac{1}{2})\pi z}{L}$ 展开，比较系数，有

$$A_{np}=\frac{2}{\rho_0^2[J_1(x_n^{(0)})]^2}\cdot\frac{2}{L}\int_0^{\rho_0}\int_0^L(u_0-u_1z^2)J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)\sin\frac{(p+\frac{1}{2})\pi z}{L}\rho d\rho dz。$$

其中积分

$$(1)\quad \int_0^L J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)\rho d\rho=\frac{\rho_0^2}{(x_n^{(0)})^2}\left[\frac{x_n^{(0)}}{\rho_0}\rho J_1\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)\right]_0^{\rho_0}=\frac{\rho_0^2}{x_n^{(0)}}J_1\left(x_n^{(0)}\right)$$

$$(2)\quad \int_0^L z^2\sin\frac{(p+\frac{1}{2})\pi z}{L}dz=\left[-\frac{L}{(p+\frac{1}{2})\pi}\cdot z^2\cdot\cos\frac{(p+\frac{1}{2})\pi z}{L}+\frac{2L^2}{(p+\frac{1}{2})^2\pi^2}\cdot z\cdot\sin\frac{(p+\frac{1}{2})\pi z}{L}+\frac{2L^3}{(p+\frac{1}{2})^3\pi^3}\cos\frac{(p+\frac{1}{2})\pi z}{L}\right]_0^L$$

$$=\frac{(-1)^p2L^2}{(p+\frac{1}{2})^2\pi^2}-\frac{2L^3}{(p+\frac{1}{2})^3\pi^3}。$$

$$(3)\quad \int_0^L \sin\frac{(p+\frac{1}{2})\pi z}{L}dz=\frac{L}{(p+\frac{1}{2})\pi}\left[-\cos\frac{(p+\frac{1}{2})\pi z}{L}\right]_0^L=\frac{L}{(p+\frac{1}{2})\pi}。$$

于是，

$$A_{np}=\frac{4}{(p+\frac{1}{2})\pi x_n^{(0)}J_1(x_n^{(0)})}\left[u_0+(-1)^{p+1}\frac{2u_1L^2}{(p+\frac{1}{2})\pi}+\frac{2u_1L^2}{(p+\frac{1}{2})^2\pi^2}\right]。$$

从而

$$X(\rho, z, t) = \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{4}{(p + \frac{1}{2})\pi x_n^{(0)} J_1(x_n^{(0)})} \left[u_0 + (-1)^{p+1} \frac{2u_1 L^2}{(p + \frac{1}{2})\pi} + \frac{2u_1 L^2}{(p + \frac{1}{2})^2 \pi^2} \right] \\ \times \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t \right\} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \sin \frac{(p + \frac{1}{2})\pi}{L} z。$$

再解 Y , 用冲量定理法, 令

$$Y(\rho, z, t) = \int_0^t w(\rho, z, t; \tau) d\tau, \quad (6)$$

w 满足定解问题

$$\begin{cases} w_t - a^2 \Delta w = 0, \\ w|_{\rho=\rho_0} = 0, \quad w|_{\rho=0} < \infty, \\ w|_{z=0} = 0, \quad \frac{\partial w}{\partial z}|_{z=L} = 0, \\ w|_{t=\tau} = 2u_1。 \end{cases} \quad (7)$$

⑦之一般解同 X , 仅需将 $t \rightarrow t - \tau$

$$w(\rho, z, t; \tau) = \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} B_{np} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \sin \frac{(p + \frac{1}{2})\pi z}{L} \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 (t - \tau) \right\}$$

将此代入⑦中初值, 类似上面计算系数, 有

$$B_{np} = - \frac{8u_1}{x_n^{(0)} (p + \frac{1}{2}) \pi J_1(x_n^{(0)})},$$

从而

$$w(\rho, z, t; \tau) = \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{-8u_1}{x_n^{(0)} (p + \frac{1}{2}) \pi J_1(x_n^{(0)})} J_0 \left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \sin \frac{(p + \frac{1}{2})\pi z}{L} \\ \times \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 (t - \tau) \right\} \quad (8)$$

将⑧代入⑥, 完成积分, 有

$$\begin{aligned}
Y(\rho, z, t) &= \int_0^t \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{-8u_1}{x_n^{(0)} (p + \frac{1}{2}) \pi J_1(x_n^{(0)})} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2}) \pi z}{L} \\
&\quad \times \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t \right\} dt \\
&= \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{8u_1}{(p + \frac{1}{2}) \pi a^2 x_n^{(0)} J_1(x_n^{(0)})} \cdot \frac{1}{\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2}} \\
&\quad \times \left\{ 1 - e^{- \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t} \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2}) \pi z}{L} z_0.
\end{aligned}$$

最终

$$\begin{aligned}
u(\rho, z, t) &= u_1 z^2 + \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{4}{(p + \frac{1}{2}) \pi x_n^{(0)} J_1(x_n^{(0)})} \left[u_0 + (-1)^{p+1} \frac{2u_1 L^2}{(p + \frac{1}{2}) \pi} + \frac{2u_1 L^2}{(p + \frac{1}{2})^2 \pi^2} \right] \\
&\quad \times \exp \left\{ - \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2}) \pi z}{L} z \\
&\quad + \sum_{n=1}^{\infty} \sum_{p=0}^{\infty} \frac{8u_1}{(p + \frac{1}{2}) \pi x_n^{(0)} a^2 J_1(x_n^{(0)})} \cdot \frac{1}{\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2}} \\
&\quad \times \left\{ 1 - e^{- \left[\left(\frac{x_n^{(0)}}{\rho_0} \right)^2 + \frac{(p + \frac{1}{2})^2 \pi^2}{L^2} \right] a^2 t} \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0} \rho\right) \sin \frac{(p + \frac{1}{2}) \pi z}{L} z_0.
\end{aligned}$$

14. 环形膜内、外半径分别是 ρ_1 和 ρ_2 ，内外边缘都保持固定,试求它的本征振动和本征频率。

解：定解问题是

$$\begin{cases} u_{tt} - a^2(u_{\rho\rho} + \frac{1}{\rho}u_{\rho} + \frac{1}{\rho^2}u_{\varphi\varphi}) = 0, \\ u|_{\rho=\rho_1} = 0, \quad u|_{\rho=\rho_2} = 0, \\ u(\rho, \varphi + 2\pi) = u(\rho, \varphi). \end{cases} \quad (1)$$

令 $u(\rho, t) = T(t)R(\rho)\Phi(\varphi)$ ，代入定解问题①，分离变数，得

$$T'' + k^2 a^2 T = 0, \quad (2)$$

以及本征问题

$$(1) \quad \begin{cases} \Phi'' + \lambda \Phi = 0 \\ \Phi(\varphi) = \Phi(\varphi + 2\pi) \end{cases} \quad (3)$$

其本征函数与本征值分别是

$$\Phi = A_m \cos m\varphi + B_m \sin m\varphi, \quad \lambda = m^2, \quad m = 0, 1, 2, \dots$$

$$(2) \quad \begin{cases} R'' + \frac{1}{\rho}R' + (k^2 - \frac{m^2}{\rho^2})R = 0, \\ R|_{\rho=\rho_1} = 0, \quad R|_{\rho=\rho_2} = 0. \end{cases} \quad (4)$$

④的一般解是 $R_n(\rho) = AJ_m(k\rho) + BN_m(k\rho)$ ，以此代入⑤中边值确定系数,有

$$\begin{cases} AJ_m(k\rho_1) + BN_m(k\rho_1) = 0, \\ AJ_m(k\rho_2) + BN_m(k\rho_2) = 0. \end{cases}$$

上式中 A 、 B 有非零解得条件是 $\begin{vmatrix} J_m(k\rho_1) & N_m(k\rho_1) \\ J_m(k\rho_2) & N_m(k\rho_2) \end{vmatrix} = 0$ 即

$$J_m(k\rho_1)N_m(k\rho_2) - J_m(k\rho_2)N_m(k\rho_1) = 0.$$

其第 n 个根记为 $k_n^{(m)}$ 。方程②的解是 $T(t) = C_n \cos k_n^{(m)}at + D_n \sin k_n^{(m)}at$

本征圆频率: $\omega_n^{(m)} = ak_n^{(m)}$,

本征振动:

$$u_n^{(m)}(\rho, \varphi, t) = \left[J_m(k_n^{(m)}\rho) - \frac{J_m(k_n^{(m)}\rho_1)}{N_m(k_n^{(m)}\rho_1)} N_m(k_n^{(m)}\rho) \right] \times (A \cos m\varphi + B \sin m\varphi)(C \cos ak_n^{(m)}t + D \sin ak_n^{(m)}t).$$

15. 半径为 ρ_0 的长圆柱面, 其径向速度分布为

$$v = v_0 \cos \varphi \cos \omega t。$$

试求解这个长圆柱面在空气中辐射出去的声场中的速度势。设 ρ_0 远小于声波波长 λ 。

解: 这是无初值问题, 其定解问题为

$$\begin{cases} u_{rr} - a^2 \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} \right) = 0 \\ \frac{\partial u}{\partial \rho} \Big|_{\rho=\rho_0} = v_0 \cos \varphi e^{-i\omega t} \end{cases} \quad (1)$$

其中, 为了方便求解, 将边值中的 $\cos \omega t$ 改写为 $e^{i\omega t}$, 在结果中取实部就是原问题的解。设①的解为 $u(\rho, t) = R(\rho) \cos \varphi e^{-i\omega t}$, 代入定解问题①, 得到

$$\begin{cases} \frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dR}{d\rho} \right) + \left(\frac{\omega^2}{a^2} - \frac{1}{\rho^2} \right) R = 0, \\ \frac{dR(\rho)}{d\rho} \Big|_{\rho=\rho_0} = v_0。 \end{cases} \quad (2)$$

这是一阶贝塞尔方程边值问题, 其一般解为

$$R(\rho) = AH_1^{(1)} \left(\frac{\omega}{a} \rho \right) + BH_1^{(2)} \left(\frac{\omega}{a} \rho \right) \quad (3)$$

对于向外辐射的柱面波, 其形式为 $e^{i(\omega\rho/a - \omega t)}$ 。考虑到大宗量时的行为, 在③中置 $B=0$, 而取

$$R(\rho) = AH_1^{(1)} \left(\frac{\omega}{a} \rho \right) \quad (4)$$

将④代入②中边界条件, 以确定系数 A

$$A \left[\frac{\partial}{\partial \rho} H_1^{(1)} \left(\frac{\omega}{a} \rho \right) \right] \Big|_{\rho=\rho_0} = v_0 \quad (5)$$

所以, $\rho_0 \ll \lambda = 2\pi/k = 2\pi a/\omega$, $\omega\rho_0/a \ll 1$, 此时

$$H_1^{(1)}(x) = J_1(x) + iN_1(x) \approx -i \frac{1}{\pi} \frac{2}{x},$$

从而, ⑤可近似为

$$A \left[\frac{\partial}{\partial \rho} \left(-i \frac{1}{\pi} \frac{2}{k\rho} \right) \right] \Big|_{\rho=\rho_0} = v_0, \Rightarrow iA \frac{2}{k\pi\rho_0^2} = v_0。$$

这样, 求得系数 $A = -i \frac{\pi v_0 \omega \rho_0^2}{2a}$ 。于是, 声场的分布为

$$u = -i \frac{\pi v_0 \omega \rho_0^2}{2a} H_1^{(1)} \left(\frac{\omega}{a} \rho \right) \cos \varphi e^{-i\omega t}。 \quad (6)$$

在远场区, 即 $\rho \left(\frac{\omega}{a} \rho \gg 1 \right)$ 大的区域, ⑥可近似为

$$u \sim -i v_0 \rho_0^2 \sqrt{\frac{\pi \omega}{2a\rho}} \cos \varphi \exp \left[i \left(\frac{\omega}{a} \rho - \omega t - 3\pi/4 \right) \right]$$

取实部

$$u \sim v_0 \rho_0^2 \sqrt{\frac{\pi \omega}{2a\rho}} \cos \varphi \cos \left[\frac{\omega}{a} (\rho - at) + \frac{3}{4} \pi \right]。$$

16. 半径为 ρ_0 的长圆柱面上一条母线作谐振动, 即柱面径向速度为 $v = v_0 \delta(\varphi - \varphi_0) \cos \omega t$ 。试求解这个长圆柱在空气中辐射出去的声场中的速度势, 设 $\rho_0 \ll \lambda$ (声波波长)。

解: 与上题类似, 定解问题是

$$\begin{cases} u_{tt} - a^2 \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} \right) = 0 \\ \left. \frac{\partial u}{\partial \rho} \right|_{\rho=\rho_0} = v_0 \delta(\varphi - \varphi_0) e^{-i\omega t} \end{cases} \quad (1)$$

设①的解为 $u(\rho, t) = R(\rho)\Phi(\varphi)e^{-i\omega t}$, 代入定解问题①, 分离变数, 得到

(1) 本征值问题

$$\begin{cases} \Phi'' + \lambda \Phi = 0 \\ \Phi(\varphi + 2\pi) = \Phi(\varphi). \end{cases} \quad (2)$$

本征函数是 $\Phi(\varphi) = C \cos m\varphi + D \sin m\varphi$, 本征值是 $\lambda = m^2$ ($m = 0, 1, 2, \dots$)

(2) 径向方程

$$\frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dR}{d\rho} \right) + \left(\frac{\omega^2}{a^2} - \frac{m^2}{\rho^2} \right) R = 0, \quad (3)$$

这是 m 阶贝塞尔方程, 其解为

$$R(\rho) = A H_m^{(1)} \left(\frac{\omega}{a} \rho \right) + B H_m^{(2)} \left(\frac{\omega}{a} \rho \right) \quad (4)$$

对于向外辐射的柱面波, 其形式为 $e^{i(\omega\rho/a - \omega t)}$, 考虑到大宗量时的行为, 在④中置 $B=0$, 而取

$$R(\rho) = A H_m^{(1)} \left(\frac{\omega}{a} \rho \right).$$

这样, 得到问题①的一般解是

$$u(\rho, \varphi, t) = \sum_{m=0}^{\infty} (C \cos m\varphi + D \sin m\varphi) H_m^{(1)} \left(\frac{\omega}{a} \rho \right) e^{-i\omega t} \quad (5)$$

将⑤代入①中边界条件

$$\sum_{m=0}^{\infty} (C \cos m\varphi + D \sin m\varphi) \left[\frac{d}{d\rho} H_m^{(1)} \left(\frac{\omega}{a} \rho \right) \right] \bigg|_{\rho=\rho_0} = v_0 \delta(\varphi - \varphi_0) \quad (6)$$

由于 $\rho_0 \ll \lambda = 2\pi/k = 2\pi a/\omega$, $\omega\rho_0/a \ll 1$, 从而有

$$\begin{aligned} (1) \quad H_m^{(1)}(x) &= J_m(x) + iN_m(x) \approx \frac{1}{m!} \left(\frac{k\rho}{2} \right)^m - i \frac{(m-1)!}{\pi} \left(\frac{2}{k\rho} \right)^m \\ &\approx -i \frac{(m-1)!}{\pi} \left(\frac{2}{k\rho} \right)^m \end{aligned}$$

$$(2) \quad H_0^{(1)}(x) = J_0(x) + iN_0(x) \approx 1 - \frac{1}{4}(k\rho)^2 + i\frac{2}{\pi}\left(\ln \frac{k\rho}{2} + C\right) \\ \approx i\frac{2}{\pi}\left(\ln \frac{k\rho}{2} + C\right)$$

$$(3) \quad \frac{d}{d\rho} H_m^{(1)}(k\rho) \approx \frac{d}{d\rho} \left[-i \frac{(m-1)!}{\pi} \left(\frac{2}{k\rho} \right)^m \right]_{\rho=\rho_0} = -i \frac{m!}{\pi} \left(\frac{2}{k} \right)^m \frac{2}{k\rho_0^{m+1}}, \quad (7)$$

$$(4) \quad \frac{d}{d\rho} H_0^{(1)}(k\rho) \Big|_{\rho=\rho_0} \approx \frac{d}{d\rho} \left[i \frac{2}{\pi} \left(\ln \frac{k\rho}{2} + C \right) \right]_{\rho=\rho_0} = i \frac{2}{\pi\rho_0}, \quad (8)$$

$\delta(\varphi - \varphi_0)$ 在 $[0, 2\pi]$ 展为傅里叶级数

$$\delta(\varphi - \varphi_0) = \frac{1}{2\pi} + \sum_{m=0}^{\infty} (\cos m\varphi_0 \cos m\varphi + \sin m\varphi_0 \sin m\varphi), \quad (9)$$

将⑦⑧⑨三式代入⑥式, 有

$$A \left(i \frac{2}{\pi\rho_0} \right) + \sum_{m=0}^{\infty} (C \cos m\varphi + D \sin m\varphi) \left[i \frac{m!}{\pi} \left(\frac{2}{k} \right)^m \frac{1}{\rho_0^{m+1}} \right] \\ = v_0 \left[\frac{1}{2\pi} + \sum_{m=0}^{\infty} (\cos m\varphi_0 \cos m\varphi + \sin m\varphi_0 \sin m\varphi) \right]$$

比较两边系数得

$$A_0 = -i \frac{v_0 \rho_0}{4},$$

$$B_0 = 0,$$

$$A_m = -i \frac{v_0 \rho_0^{m+1} k^m}{2^m m!} \cos m\varphi_0,$$

$$B_m = -i \frac{v_0 \rho_0^{m+1} k^m}{2^m m!} \sin m\varphi_0,$$

于是, 声场的分布为

$$u(\rho, \varphi, t) = \operatorname{Re} \left\{ \left[\frac{v_0 \rho_0}{4} H_0^{(1)}(k\rho) + \sum_{m=1}^{\infty} H_m^{(1)}(k\rho) \right. \right. \\ \left. \left. \times \frac{v_0 k^m \rho_0^{m+1}}{2^m m!} (\cos m\varphi_0 \cos m\varphi + \sin m\varphi_0 \sin m\varphi) \right] e^{-i(\omega t + \frac{\pi}{2})} \right\}.$$

17. 在半径为 ρ_0 的圆盘区域内部求解热传导方程定解问题

$$\begin{cases} u_t - a^2 \Delta u = 0, \\ u|_{\rho=\rho_0} = 0, \\ u|_{t=0} = u_0 \sin \pi \rho / \rho_0. \end{cases} \quad (1)$$

解：类似第 10 题，①之解的形式为 $J_0(k\rho)e^{-k^2 a^2 t}$ ，这里， $k\rho_0 = x_n^{(0)}$ [$x_n^{(0)}$ 为 $J_0(x)=0$ 的根]。于是，①的一般解为

$$v(\rho, t) = \sum_{n=1}^{\infty} A_n J_0 \left[\left(\frac{x_n^{(0)}}{\rho_0} \rho \right) \right] \exp \left\{ - \left[\frac{x_n^{(0)}}{\rho_0} \right]^2 a^2 t \right\} \quad (2)$$

将②代入①中初值，

$$\sum_{n=1}^{\infty} A_n J_0 \left[\frac{x_n^{(0)}}{\rho_0} \rho \right] = u_0 \sin \frac{\pi \rho}{\rho_0}。$$

有

$$A_n = \frac{2 \int_0^{\rho_0} \xi J_0 \left[\frac{x_n^{(0)}}{\rho_0} \xi \right] \sin \frac{\pi \xi}{\rho_0} d\xi}{\rho_0^2 J_1^2 \left[x_n^{(0)} \right]}。$$

解为

$$u(\rho, t) = \sum_{n=1}^{\infty} \frac{2 \int_0^{\rho_0} \xi J_0 \left[\frac{x_n^{(0)}}{\rho_0} \xi \right] \sin \frac{\pi \xi}{\rho_0} d\xi}{\rho_0^2 J_1^2 \left[x_n^{(0)} \right]} J_0 \left[\frac{x_n^{(0)}}{\rho_0} \rho \right] \exp \left\{ - \left[\frac{x_n^{(0)}}{\rho_0} \right]^2 a^2 t \right\}。$$

18. 已知半径为 ρ_0 的圆盘, 其边界自由, 初始位移为 $A_0\rho\cos\varphi$, 由静止状态释放. 试求解圆盘内部振动方程 $u_{tt} - a^2\Delta u = 0$ 的定解问题.

解: 定解问题为

$$\begin{cases} u_{tt} - a^2\Delta u = 0, \\ u|_{\rho=\rho_0} = 0, \\ u|_{t=0} = A_0\rho\cos\varphi, \\ u_t|_{t=0} = 0. \end{cases} \quad (1)$$

令 $u(\rho, \varphi, t) = T(t)R(\rho, \varphi)$, 代入定解问题①, 分离变数, 得

$$T'' + k^2 a^2 T = 0, \quad (2)$$

及本征值问题

$$(1) \quad \begin{cases} \Phi'' + \lambda\Phi = 0 \\ \Phi(\varphi + 2\pi) = \Phi(\varphi). \end{cases}$$

本征函数是 $\Phi(\varphi) = C\cos m\varphi + D\sin m\varphi$, 本征值是 $\lambda = m^2$ ($m = 0, 1, 2, \dots$). 根据初始条件, 取 $m=1, D=0$, 从而, $\Phi(\varphi) = C\cos\varphi$,

$$(2) \quad \begin{cases} R'' + \frac{1}{\rho}R' + \left(k^2 - \frac{1}{\rho^2}\right)R = 0 \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty, \end{cases} \Rightarrow \begin{cases} \rho^2 R'' + \rho R' + (k^2 \rho^2 - 1)R = 0 \\ R|_{\rho=\rho_0} = 0, \quad R|_{\rho=0} < \infty, \end{cases} \quad (3)$$

③的本征函数是 $R_n(\rho) = J_1(k_n^{(1,2)}\rho)$, 本征值是 $k_n^{(1,2)} = \frac{x_n^{(1,2)}}{\rho_0}$, $n=1, 2, \dots$. 其中 $x_n^{(1,2)}$ 为

$J_1'(x)=0$ 的第 n 个根方程②的解是 $T(t) = A_n \cos k_n^{(1,2)}at + B_n \sin k_n^{(1,2)}at$, 于是, 问题①的一般解是

$$u(\rho, \varphi, t) = \sum_{n=1}^{\infty} (A_n \cos k_n^{(1,2)}at + B_n \sin k_n^{(1,2)}at) J_1(k_n^{(1,2)}\rho) \cos\varphi \quad (3)$$

将③代入①中初始条件以确定系数, 由 $u_t|_{t=0} = 0$, 有 $B_n = 0$.

由 $u|_{t=0} = \sum_{n=1}^{\infty} A_n J_1(k_n^{(1,2)}\rho) \cos\varphi = A_0\rho\cos\varphi$, 有

$$A_n = \frac{A_0}{N^2} \int_0^{\rho_0} J_1\left(\frac{x_n^{(1,2)}}{\rho_0}\rho\right) \rho^2 d\rho.$$

其中 $\int_0^{\rho_0} J_1\left(\frac{x_n^{(1,2)}}{\rho_0}\rho\right) \rho^2 d\rho = \frac{\rho_0^3}{(x_n^{(1,2)})^3} \int_0^{x_n^{(1,2)}} x^2 J_1(x) dx = \frac{\rho_0^3}{x_n^{(1,2)}} J_2(x_n^{(1,2)}),$

$$N^2 = \int_0^{\rho_0} J_1^2\left(\frac{x_n^{(1,2)}}{\rho_0}\rho\right) \rho d\rho = \frac{1}{2} \left[\rho_0^2 - \frac{\rho_0^2}{(x_n^{(1,2)})^2} \right] [J_1(x_n^{(1)})]^2.$$

从而

$$A_n = \frac{2A_0 x_n^{(1,2)} \rho_0 J_2(x_n^{(1,2)})}{\left[(x_n^{(1,2)})^2 - 1\right] J_1^2(x_n^{(1,2)})}.$$

解为

$$u(\rho, \varphi, t) = \sum_{n=1}^{\infty} \frac{2A_0 x_n^{(1,2)} \rho_0 J_2(x_n^{(1,2)})}{\left[(x_n^{(1,2)})^2 - 1\right] J_1^2(x_n^{(1,2)})} J_1\left(\frac{x_n^{(1,2)}}{\rho_0}\rho\right) \cos\varphi \cos \frac{x_n^{(1,2)}}{\rho_0} at.$$

§11. 4 虚宗量贝塞尔方程

1. 试证明虚宗量贝塞尔函数 $I_\nu(x)$ 下列递推关系

$$(1) \quad \frac{d}{dx} \left[\frac{I_\nu(x)}{x^\nu} \right] = \frac{I_{\nu+1}(x)}{x^\nu}; \quad \frac{d}{dx} [x^\nu I_\nu(x)] = x^\nu I_{\nu-1}(x).$$

证: i) 由

$$\begin{aligned} \frac{d}{dx} \left[\frac{I_\nu(x)}{x^\nu} \right] &= \frac{d}{dx} \left[\frac{1}{x^\nu} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{\nu+2k} \right] \\ &= \frac{d}{dx} \left[\sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{2k} \right] \\ &= \sum_{k=0}^{\infty} \frac{2k}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{2k-1} \end{aligned}$$

将 k 换成 $k+1$

$$\begin{aligned} \frac{d}{dx} \left[\frac{I_\nu(x)}{x^\nu} \right] &= \sum_{k=0}^{\infty} \frac{2(k+1)}{(k+1)! \Gamma(\nu+k+1+1)} \left(\frac{1}{2} \right)^{\nu+2(k+1)} x^{2(k+1)-1} \\ &= \frac{1}{x^\nu} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu+1+k+1)} \left(\frac{1}{2} \right)^{\nu+1+2k} x^{\nu+1+2k} \\ &= \frac{I_{\nu+1}(x)}{x^\nu} \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{d}{dx} [x^\nu I_\nu(x)] &= \frac{d}{dx} \left[x^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{\nu+2k} \right] \\ &= \sum_{k=0}^{\infty} \frac{2\nu+2k}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{2\nu+2k-1} \\ &= x^\nu \sum_{k=0}^{\infty} \frac{2(\nu+k)}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{\nu+2k-1} \end{aligned}$$

利用 Γ 函数的性质: $\Gamma(x+1) = x\Gamma(x)$, 有

$$\begin{aligned} \frac{d}{dx} [x^\nu I_\nu(x)] &= x^\nu \sum_{k=0}^{\infty} \frac{2(\nu+k)}{k! \Gamma(\nu+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{\nu+2k-1} \\ &= x^\nu \sum_{k=0}^{\infty} \frac{2(\nu+k)}{k!(\nu+k)\Gamma(\nu-1+k+1)} \left(\frac{1}{2} \right)^{\nu+2k} x^{\nu+2k-1} \\ &= x^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu-1+k+1)} \left(\frac{1}{2} \right)^{\nu-1+2k} x^{\nu-1+2k} \\ &= x^\nu I_{\nu-1}(x) \end{aligned}$$

证毕。

$$(2) \quad I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu I_\nu(x)}{x}; \quad I_{\nu-1}(x) + I_{\nu+1}(x) = 2I'_\nu(x).$$

证: 将 (1) 中两式左端展开, 有

$$I'_\nu(x) - \frac{\nu I_\nu(x)}{x} = I_{\nu+1}(x); \tag{①}$$

$$I'_\nu(x) + \frac{\nu I_\nu(x)}{x} = I_{\nu-1}(x); \tag{②}$$

②-①, 得

$$I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu I_\nu(x)}{x};$$

②+①, 得

$$I_{\nu-1}(x) + I_{\nu+1}(x) = 2I'_\nu(x),$$

证毕。

2. 试证明虚宗量汉克函数 $K_\nu(x)$ 下列递推关系

$$(1) \quad \frac{d}{dx} \left[\frac{K_\nu(x)}{x^\nu} \right] = -\frac{K_{\nu+1}(x)}{x^\nu}; \quad \frac{d}{dx} [x^\nu K_\nu(x)] = -x^\nu K_{\nu-1}(x).$$

证: 由 $K_\nu(x) = \frac{\pi}{2} \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu \pi}$ 。

i) 类似于第 1 题, 可求得 $\frac{d}{dx} \left[\frac{I_{-\nu}(x)}{x^\nu} \right] = \frac{I_{-\nu-1}(x)}{x^\nu}$ 。

$$\begin{aligned} \frac{d}{dx} \left[\frac{K_\nu(x)}{x^\nu} \right] &= \frac{d}{dx} \left[\frac{\pi}{2} \frac{I_{-\nu}(x) - I_\nu(x)}{x^\nu \sin \nu \pi} \right] \\ &= \frac{\pi}{2 \sin \nu \pi} \frac{d}{dx} \left[\frac{I_{-\nu}(x) - I_\nu(x)}{x^\nu} \right] \\ &= \frac{\pi}{2 \sin \nu \pi} \left[\frac{I_{-\nu-1}(x) - I_{\nu+1}(x)}{x^\nu} \right] \\ &= \frac{\sin(\nu+1)\pi}{x^\nu \sin \nu \pi} \left[\frac{\pi}{2} \frac{I_{-(\nu+1)}(x) - I_{\nu+1}(x)}{\sin(\nu+1)\pi} \right] \end{aligned} \quad (1)$$

其中 $\sin(\nu+1)\pi = -\sin \nu \pi$ 。从而 (1) 式为

$$\frac{d}{dx} \left[\frac{K_\nu(x)}{x^\nu} \right] = -\frac{1}{x^\nu} \left[\frac{\pi}{2} \frac{I_{-(\nu+1)}(x) - I_{\nu+1}(x)}{\sin(\nu+1)\pi} \right] = -\frac{K_{\nu+1}(x)}{x^\nu}。$$

ii) 类似于第 1 题, 可求得 $\frac{d}{dx} [x^\nu I_{-\nu}(x)] = x^\nu I_{-\nu+1}(x)$ 。

$$\begin{aligned} \frac{d}{dx} [x^\nu K_\nu(x)] &= \frac{d}{dx} \left[x^\nu \cdot \frac{\pi}{2} \cdot \frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu \pi} \right] \\ &= \frac{\pi}{2 \sin \nu \pi} \frac{d}{dx} \{ x^\nu [I_{-\nu}(x) - I_\nu(x)] \} \\ &= \frac{\pi}{2 \sin \nu \pi} [x^\nu I_{-\nu+1}(x) - x^\nu I_{\nu-1}(x)] \\ &= x^\nu \frac{\sin(\nu-1)\pi}{\sin \nu \pi} \left[\frac{\pi}{2} \frac{I_{-(\nu-1)}(x) - x^\nu I_{\nu-1}(x)}{\sin(\nu-1)\pi} \right] \\ &= -x^\nu K_{\nu-1}(x)。 \end{aligned}$$

证毕。

$$(2) \quad K_{\nu-1}(x) - K_{\nu+1}(x) = -\frac{2\nu K_\nu(x)}{x}; \quad K_{\nu-1}(x) + K_{\nu+1}(x) = -2K'_\nu(x).$$

证: 将 (1) 中两式左端展开, 有

$$K'_\nu(x) - \frac{\nu K_\nu(x)}{x} = -K_{\nu+1}(x), \quad (2)$$

$$K'_\nu(x) + \frac{\nu K_\nu(x)}{x} = -K_{\nu-1}(x)。 \quad (3)$$

(2)-(3), 得

$$K_{\nu-1}(x) - K_{\nu+1}(x) = -\frac{2\nu K_\nu(x)}{x}。$$

(2)+(3), 得

$$K_{\nu-1}(x) + K_{\nu+1}(x) = -2K'_\nu(x)。$$

证毕。

3. 匀质圆柱半径为 ρ_0 、高为 L 。下底保持温度 u_1 ，上底保持温度 u_2 。侧面温度分布为 $f(z) = \frac{2u_2}{L^2}(z - \frac{L}{2})z + \frac{u_1}{L}(L - z)$ 。求解柱体内各点的稳恒温度。

解：取柱坐标系，定解问题为

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = \frac{2u_2}{L^2} \left(z - \frac{L}{2} \right) z + \frac{u_1}{L} (L - z), \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = u_1, \quad u|_{z=L} = u_2. \end{cases} \quad (1)$$

边界条件全为非齐次，需做齐次化处理，上下底面条件相对简单，齐次化上下底面条件。令

$u(\rho, z) = u_1 + \frac{u_2 - u_1}{L}z + v$ 代入①，得 v 的定解问题

$$\begin{cases} \Delta v = 0, \\ v|_{\rho=\rho_0} = \frac{2u_2}{L^2}z^2 - \frac{2u_2}{L}z, \quad v|_{\rho=0} < \infty, \\ v|_{z=0} = 0, \quad v|_{z=L} = 0. \end{cases} \quad (2)$$

②分离变数后，由 Z 向方程及对应齐次边条件构成本征问题

$$\begin{cases} Z'' - \mu Z = 0, \\ Z|_{z=0} = 0, \quad Z|_{z=L} = 0. \end{cases}$$

本征函数是 $\sin \frac{n\pi z}{L}$ ，本征值 $-\mu = \frac{n^2\pi^2}{L^2}$ ， $n=1, 2, \dots$ 。径向为虚宗量贝塞尔方程，柱内有限解为虚宗量贝塞尔函数，这样，问题的一般解为

$$v(\rho, z) = \sum_{n=1}^{\infty} A_n I_0 \left(\frac{n\pi}{L} \rho \right) \sin \frac{n\pi z}{L} \quad (3)$$

将③代入②中径向边界条件，以确定系数

$$\sum_{n=1}^{\infty} A_n I_0 \left(\frac{n\pi}{L} \rho_0 \right) \sin \frac{n\pi z}{L} = \frac{2u_2}{L^2} z^2 - \frac{2u_2}{L} z,$$

有

$$\begin{aligned} A_n &= \frac{2}{L} \cdot \frac{1}{I_0 \left(\frac{n\pi}{L} \rho_0 \right)} \int_0^L \left(\frac{2u_2}{L^2} z^2 - \frac{2u_2}{L} z \right) \sin \frac{n\pi z}{L} dz \\ &= \frac{4u_2}{L^2} \cdot \frac{1}{I_0 \left(\frac{n\pi}{L} \rho_0 \right)} \int_0^L \left(\frac{1}{L} z^2 - z \right) \sin \frac{n\pi z}{L} dz, \end{aligned}$$

其中积分

$$\begin{aligned} (1) \quad \frac{1}{L} \int_0^L z^2 \sin \frac{n\pi z}{L} dz &= \frac{1}{L} \left(\frac{2L^2}{n^2\pi^2} \cdot z \cdot \sin \frac{n\pi z}{L} + \frac{2L^3}{n^3\pi^3} \cos \frac{n\pi z}{L} - \frac{L}{n\pi} \cdot z^2 \cdot \cos \frac{n\pi z}{L} \right) \Big|_0^L \\ &= [(-1)^{n+1} \frac{L^2}{n\pi} + (-1)^n \frac{2L^2}{n^3\pi^3}] - \frac{2L^2}{n^3\pi^3} \\ &= (-1)^{n+1} \frac{L^2}{n\pi} + \frac{2L^2}{n^3\pi^3} [(-1)^n - 1], \end{aligned}$$

$$(2) \quad \int_0^L z \sin \frac{n\pi z}{L} dz = \left(\frac{L^2}{n^2\pi^2} \sin \frac{n\pi z}{L} - \frac{Lz}{n\pi} \cos \frac{n\pi z}{L} \right) \Big|_0^L = (-1)^{n+1} \frac{L^2}{n\pi}.$$

这样

$$A_n = \frac{1}{I_0 \left(\frac{n\pi}{L} \rho_0 \right)} \frac{8u_2}{n^3\pi^3} [(-1)^n - 1] = \begin{cases} \frac{1}{I_0 \left(\frac{(2k+1)\pi}{L} \rho_0 \right)} \frac{16u_2}{(2k+1)^3\pi^3}, & (n=2k+1), \\ 0, & (n=2k), \end{cases}$$

其中， $k=0, 1, 2, \dots$ 。最终，问题的解是

$$\begin{aligned} u(\rho, z) &= u_1 + \frac{u_2 - u_1}{L}z - \sum_{k=0}^{\infty} \frac{16u_2}{(2k+1)^3\pi^3} \left[I_0 \left(\frac{(2k+1)\pi}{L} \rho \right) / I_0 \left(\frac{(2k+1)\pi}{L} \rho_0 \right) \right] \\ &\quad \times \sin \frac{(2k+1)\pi}{L} z. \end{aligned}$$

4. 匀质圆柱半径为 ρ_0 、高为 L 。上底有均匀分布的强度为 q_0 的热流进入，下底保持温度 u_0 ，侧面温度分布为 $f(z)$ ，求解柱体内各点的稳恒温度。

解：取柱坐标系，定解问题为

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = f(z), \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = u_0, \quad u_z|_{z=L} = \frac{q_0}{k}. \end{cases} \quad (1)$$

边界条件全为非齐次，需做齐次化处理，上下底面条件相对简单，齐次化上下底面条件。令

$u(\rho, z) = v(\rho, z) + u_0 + \frac{q_0}{k}z$ 代入①，转换成 v 的定解问题

$$\begin{cases} \Delta v = 0, \\ v|_{\rho=\rho_0} = f(z) - u_0 - \frac{q_0}{k}z, \quad v|_{\rho=0} < \infty, \\ v|_{z=0} = 0, \quad v_z|_{z=L} = 0. \end{cases} \quad (2)$$

分离变数后，由 Z 向方程及对应齐次边条件构成本征问题

$$\begin{cases} Z'' - \mu Z = 0, \\ Z|_{z=0} = 0, \quad Z_z|_{z=L} = 0. \end{cases}$$

本征函数是 $\sin \frac{(p+\frac{1}{2})\pi z}{L}$ ，本征值 $-\mu = \frac{(p+\frac{1}{2})^2 \pi^2}{L^2}$ ， $p = 0, 1, 2, \dots$ 。径向为虚宗量贝塞尔方程，柱内有限解为虚宗量贝塞尔函数，这样，②的一般解为

$$v(\rho, z) = \sum_{n=1}^{\infty} A_n I_0 \left(\frac{(p+\frac{1}{2})\pi}{L} \rho \right) \sin \frac{(p+\frac{1}{2})\pi z}{L} \quad (3)$$

将③代入②中径向边界条件，以确定系数

$$\sum_{n=1}^{\infty} A_n I_0 \left(\frac{(p+\frac{1}{2})\pi}{L} \rho_0 \right) \sin \frac{(p+\frac{1}{2})\pi z}{L} = f(z) - u_0 - \frac{q_0}{k}z,$$

有

$$A_n = \frac{2}{L} \cdot \frac{1}{I_0 \left(\frac{(p+\frac{1}{2})\pi}{L} \rho_0 \right)} \int_0^L \left(f(\xi) - u_0 - \frac{q_0}{k}\xi \right) \sin \frac{(p+\frac{1}{2})\pi \xi}{L} d\xi.$$

其中，记

$$\frac{2}{L} \int_0^L f(\xi) \sin \frac{(p+\frac{1}{2})\pi \xi}{L} d\xi \equiv f_p,$$

为 $f(z)$ 按 $\sin \frac{(p+\frac{1}{2})\pi z}{L}$ 展开的傅里叶系数。积分

$$(1) \quad -u_0 \cdot \frac{2}{L} \int_0^L \sin \frac{(p+\frac{1}{2})\pi \xi}{L} d\xi = -\frac{2u_0}{(p+\frac{1}{2})\pi},$$

$$\begin{aligned} (2) \quad & -\frac{2}{L} \cdot \frac{q_0}{k} \int_0^L \xi \sin \frac{(p+\frac{1}{2})\pi \xi}{L} d\xi \\ &= -\frac{2}{L} \cdot \frac{q_0}{k} \left[\frac{L^2}{(p+\frac{1}{2})^2 \pi^2} \sin \frac{(p+\frac{1}{2})\pi \xi}{L} - \frac{L^2}{(p+\frac{1}{2})^2 \pi^2} \cdot \xi \cdot \cos \frac{(p+\frac{1}{2})\pi \xi}{L} \right]_0^L \\ &= (-1)^{p+1} \frac{2q_0 L}{k(p+\frac{1}{2})^2 \pi^2} \end{aligned}$$

最终

$$\begin{aligned} u(\rho, z) = & u_0 + \frac{q_0}{k} z + \sum_{p=0}^{\infty} \frac{1}{\mathrm{I}_0 \left(\frac{(p+\frac{1}{2})\pi}{L} \rho_0 \right)} \\ & \times \left\{ f_p - \frac{2u_0}{(p+\frac{1}{2})\pi} + \frac{(-1)^{p+1} 2q_0 L}{k(p+\frac{1}{2})^2 \pi^2} \right\} \mathrm{I}_0 \left(\frac{(p+\frac{1}{2})\pi}{L} \rho \right) \sin \frac{(p+\frac{1}{2})\pi}{L} z. \end{aligned}$$

其中 f_p 为 $f(z) = \sum_{p=0}^{\infty} f_p \sin \frac{(p+\frac{1}{2})\pi}{L} z$ 的傅里叶系数。

5. 如图 11-1 电子光学透镜的某一部件由两个中空圆柱筒组成, 其电势分别为 $+v_0$ 和 $-v_0$ 。在圆柱中间隙缝的边缘处电势可近似表为 $v = v_0 \sin \frac{\pi z}{2\delta}$ 。求圆柱筒内的电势分布。圆柱两端边界条件可近似表为 $v|_{z=\pm l} = \pm v_0$ 。圆柱筒的半径为 ρ_0 。

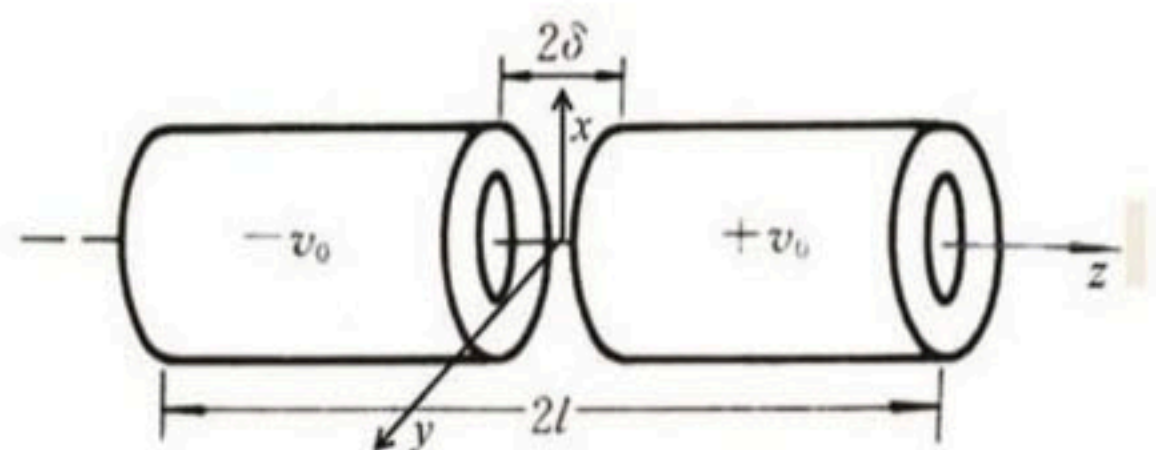


图11-1

解: 取柱坐标系如图 11-1 所示。根据题意, 在此坐标系中, 电势按 z 轴成奇对称分布, 因此, 只需在 $[0, l]$ 上求解。定解问题是

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta}, & (0 \leq z \leq \delta) \\ v_0, & (\delta \leq z \leq l) \end{cases} \\ u|_{\rho=0} < \infty, \\ u|_{z=0} = v_0 \sin \frac{\pi z}{2\delta} \Big|_{z=0} = 0, \quad u|_{z=l} = v_0. \end{cases} \quad (1)$$

由于径向与 z 向均为非齐次边值, 将 u 的定解问题①分成两个定解问题。令 $u = v + w$, 代入①, 分解得

$$(1) \quad \begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{\partial^2 v}{\partial z^2} = 0, \\ v|_{\rho=\rho_0} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta}, & (0 \leq z \leq \delta) \\ v_0, & (\delta \leq z \leq l) \end{cases} \\ v|_{\rho=0} < \infty, \\ v|_{z=0} = 0, \quad v|_{z=l} = 0. \end{cases} \quad (2)$$

$$(2) \quad \begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial w}{\partial \rho} \right) + \frac{\partial^2 w}{\partial z^2} = 0, \\ w|_{\rho=\rho_0} = 0, \\ w|_{\rho=0} < \infty, \\ w|_{z=0} = 0, \quad w|_{z=l} = v_0. \end{cases} \quad (3)$$

先解 v , 由于上下底面为第一类齐次边值, 分离变数后, 由 z 方向方程与齐次边值构成本征问题。

径向为虚宗量贝塞尔方程, 此时, 柱内有限解是

$$v(\rho, z) = \sum_{p=1}^{\infty} A_p I_0 \left(\frac{p\pi}{l} \rho \right) \sin \frac{p\pi z}{l}, \quad (4)$$

将④代入②中径向边值, 有

$$\sum_{p=1}^{\infty} A_p I_0 \left(\frac{p\pi}{l} \rho_0 \right) \sin \frac{p\pi z}{l} = \begin{cases} v_0 \sin \frac{\pi z}{2\delta}, & (0 \leq z \leq \delta) \\ v_0, & (\delta \leq z \leq l) \end{cases}$$

系数为

$$A_p = \frac{1}{I_0 \left(\frac{p\pi}{l} \rho_0 \right)} \cdot \frac{2}{l} \left[\int_0^{\delta} v_0 \sin \frac{\pi z}{2\delta} \sin \frac{p\pi z}{l} dz + \int_{\delta}^l v_0 \sin \frac{p\pi z}{l} dz \right],$$

其中积分

$$\begin{aligned} (1) \quad \int_0^{\delta} \sin \frac{\pi z}{2\delta} \sin \frac{p\pi z}{l} dz &= \left[-\frac{\sin \left(\frac{\pi}{2\delta} + \frac{p\pi}{l} \right) z}{2 \left(\frac{\pi}{2\delta} + \frac{p\pi}{l} \right)} + \frac{\sin \left(\frac{\pi}{2\delta} - \frac{p\pi}{l} \right) z}{2 \left(\frac{\pi}{2\delta} - \frac{p\pi}{l} \right)} \right]_0^{\delta} \\ &= -\frac{\sin \left(\frac{\pi}{2} + \frac{p\pi\delta}{l} \right)}{2 \left(\frac{\pi}{2\delta} + \frac{p\pi}{l} \right)} + \frac{\sin \left(\frac{\pi}{2} - \frac{p\pi\delta}{l} \right)}{2 \left(\frac{\pi}{2\delta} - \frac{p\pi}{l} \right)} \\ &= \frac{\cos \frac{p\pi\delta}{l}}{2} \cdot \frac{\left(\frac{\pi}{2\delta} + \frac{p\pi}{l} \right) - \left(\frac{\pi}{2\delta} - \frac{p\pi}{l} \right)}{\left(\frac{\pi}{2\delta} \right)^2 - \left(\frac{p\pi}{l} \right)^2} \\ &= \frac{\cos \frac{p\pi\delta}{l}}{2} \cdot \frac{2 \cdot \frac{p\pi}{l}}{\left(\frac{\pi}{2\delta} \right)^2 - \left(\frac{p\pi}{l} \right)^2} \\ &= \cos \frac{p\pi\delta}{l} \cdot \frac{p\pi}{l} \cdot \frac{4\delta^2 l^2}{l^2 \pi^2 - 4\delta^2 p^2 \pi^2} \end{aligned}$$

$$(2) \quad \int_{\delta}^l \sin \frac{p\pi z}{l} dz = \frac{l}{p\pi} \left[-\cos \frac{p\pi z}{l} \right]_{\delta}^l = \frac{l}{p\pi} \left[(-1)^{p+1} + \cos \frac{p\pi\delta}{l} \right]$$

从而

$$A_p = \frac{v_0}{I_0 \left(\frac{p\pi}{l} \rho_0 \right)} \cdot \frac{2}{l} \left\{ \frac{p\pi}{l} \cdot \frac{4\delta^2 l^2}{l^2 \pi^2 - 4\delta^2 p^2 \pi^2} \cos \frac{p\pi\delta}{l} + \frac{l}{p\pi} \left[(-1)^{p+1} + \cos \frac{p\pi\delta}{l} \right] \right\}$$

$$\begin{aligned}
&= \frac{2v_0}{I_0\left(\frac{p\pi}{l}\rho_0\right)} \cdot \frac{1}{p\pi} \left[\frac{4p^2\delta^2}{l^2-4p^2\delta^2} \cos \frac{p\pi\delta}{l} + (-1)^{p+1} + \cos \frac{p\pi\delta}{l} \right] \\
&= \frac{2v_0}{I_0\left(\frac{p\pi}{l}\rho_0\right)} \cdot \frac{1}{p\pi} \left[(-1)^{p+1} + \left(1 + \frac{4p^2\delta^2}{l^2-4p^2\delta^2}\right) \cos \frac{p\pi\delta}{l} \right] \\
&= \frac{2v_0}{\pi} \frac{1}{I_0\left(\frac{p\pi}{l}\rho_0\right)} \cdot \frac{1}{p} \left[(-1)^{p+1} + \frac{l^2}{l^2-4p^2\delta^2} \cos \frac{p\pi\delta}{l} \right]
\end{aligned}$$

故:

$$v = \frac{2v_0}{\pi} \sum_{p=1}^{\infty} \frac{1}{p} \left[(-1)^{p+1} + \frac{l^2}{l^2-4p^2\delta^2} \cos \frac{p\pi\delta}{l} \right] \frac{I_0\left(\frac{p\pi}{l}\rho\right)}{I_0\left(\frac{p\pi}{l}\rho_0\right)} \sin \frac{p\pi z}{l}.$$

再解 w , 由于侧面为第一类齐次边值, 分离变数后, 由径向贝塞尔方程与齐次边值构成本征问题。本征函数与本征值分别为 $J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)$, $\sqrt{\mu_n^{(0)}} = \frac{x_n^{(0)}}{\rho_0}$, $n=1,2,\dots$, $x_n^{(0)}$ 是零阶贝塞尔方程的第 n 个零点。此时, 柱内有限解是

$$w(\rho, z) = \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)}z/\rho_0} + B_n e^{-x_n^{(0)}z/\rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \quad (5)$$

将⑤代入③中 z 向边值, 有

$$\begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) = 0 \\ \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)}L/\rho_0} + B_n e^{-x_n^{(0)}L/\rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) = v_0. \end{cases} \quad (6)$$

类似于第 4 题, 从⑥中解出 A_n 、 B_n ,

$$\begin{cases} B_n = -A_n, \\ A_n = \frac{2}{\rho_0^2 [J_1(x_n^{(0)})]^2} \cdot \frac{1}{2\text{sh}(x_n^{(0)}l/\rho_0)} \int_0^{\rho_0} v_0 J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \rho d\rho \\ = \frac{v_0 \rho_0^2}{\rho_0^2 (x_n^{(0)})^2 [J_1(x_n^{(0)})]^2 \text{sh}(x_n^{(0)}l/\rho_0)} \left[\frac{x_n^{(0)}}{\rho_0} \rho J_1\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \right]_0^{\rho_0} \\ = \frac{v_0}{x_n^{(0)} J_1(x_n^{(0)}) \text{sh}(x_n^{(0)}l/\rho_0)}. \end{cases}$$

于是

$$w(\rho, \varphi) = 2v_0 \sum_{n=1}^{\infty} \frac{1}{x_n^{(0)} J_1(x_n^{(0)}) \text{sh}(x_n^{(0)}l/\rho_0)} J_1\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \text{sh}\left(\frac{x_n^{(0)}}{\rho_0}z\right)$$

结果

$$\begin{aligned}
u(\rho, z) &= \frac{2v_0}{\pi} \sum_{p=1}^{\infty} \frac{1}{p} \left[(-1)^{p+1} + \frac{l^2}{l^2-4p^2\delta^2} \cos \frac{p\pi\delta}{l} \right] \frac{I_0(p\pi\rho/l)}{I_0(p\pi\rho_0/l)} \sin \frac{p\pi z}{l} \\
&\quad + 2v_0 \sum_{n=1}^{\infty} \frac{1}{x_n^{(0)} J_1(x_n^{(0)})} \cdot \frac{\text{sh}(x_n^{(0)}z/\rho_0)}{\text{sh}(x_n^{(0)}l/\rho_0)} J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right).
\end{aligned}$$

6. 利用虚宗量贝塞尔函数重解 § 11.2 的习题 6。

解：根据题意，定解问题是

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = 0, \quad u|_{\rho=0} < \infty, \\ \left. \frac{\partial u}{\partial z} \right|_{z=0} = \frac{q_0}{k}, \quad \left. \frac{\partial u}{\partial z} \right|_{z=L} = \frac{q_0}{k}. \end{cases} \quad (1)$$

为用虚宗量贝塞尔函数，现将上下底面边界条件齐次化。令 $u(\rho, z) = v(\rho, z) + \frac{q_0}{k} z$ 原定解问题①转换成 v 的定解问题

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 v}{\partial \varphi^2} + \frac{\partial^2 v}{\partial z^2} = 0, \\ v|_{\rho=\rho_0} = -\frac{q_0}{k} z, \quad u|_{\rho=0} < \infty, \\ \left. \frac{\partial v}{\partial z} \right|_{z=0} = 0, \quad \left. \frac{\partial v}{\partial z} \right|_{z=L} = 0. \end{cases} \quad (2)$$

对②分离变数，由 Z 向方程及对应齐次边条件构成本征问题

$$\begin{cases} Z'' - \mu Z = 0, \\ Z_z|_{z=0} = 0, \quad Z_z|_{z=L} = 0. \end{cases}$$

本征函数是 $\cos \frac{m\pi z}{L}$ ，本征值是 $-\mu = \frac{m^2 \pi^2}{L^2}$ ， $m = 0, 1, 2, \dots$ 。径向为虚宗量贝塞尔方程，柱内有限解为虚宗量贝塞尔函数，这样，问题的一般解为

$$v(\rho, z) = \sum_{m=0}^{\infty} A_m I_0 \left(\frac{m\pi}{L} \rho \right) \cos \frac{m\pi z}{L} \quad (3)$$

将③代入②中径向边界条件，以确定系数，有

$$\sum_{m=0}^{\infty} A_m I_0 \left(\frac{m\pi}{L} \rho_0 \right) \cos \frac{m\pi z}{L} = -\frac{q_0}{k} z,$$

由此解出

$$A_m = \frac{2}{L} \cdot \frac{1}{I_0 \left(\frac{m\pi}{L} \rho_0 \right)} \int_0^L \left(-\frac{q_0}{k} \xi \right) \cos \frac{m\pi \xi}{L} d\xi$$

其中积分，

$$\begin{aligned}
& -\frac{2}{L} \cdot \frac{q_0}{k} \int_0^L \xi \cos \frac{m\pi\xi}{L} d\xi \\
& = -\frac{2}{L} \cdot \frac{q_0}{k} \left[\frac{L^2}{m^2\pi^2} \cos \frac{m\pi\xi}{L} - \frac{L}{m\pi} \cdot \xi \cdot \sin \frac{m\pi\xi}{L} \right]_0^L \\
& = \frac{2q_0L}{km^2\pi^2} [1 - (-1)^m]
\end{aligned}$$

于是

$$\begin{aligned}
A_m &= \frac{1}{I_0\left(\frac{m\pi}{L}\rho_0\right)} \cdot \frac{2q_0L}{km^2\pi^2} [1 - (-1)^m] \\
&= \begin{cases} \frac{1}{I_0\left(\frac{(2n-1)\pi}{L}\rho_0\right)} \cdot \frac{4q_0L}{k(2n-1)^2\pi^2}, & m=2n-1, \quad n=1,2,\dots \\ 0, & m=2n, \quad n=1,2,\dots \end{cases} \\
A_0 &= \frac{1}{L} \cdot \frac{1}{I_0(0)} \int_0^L \left(-\frac{q_0}{k}\xi\right) d\xi = -\frac{q_0L}{2k}
\end{aligned}$$

最终

$$\begin{aligned}
u(\rho, z) &= \frac{q_0}{k} z - \frac{q_0L}{2k} + \sum_{n=1}^{\infty} \frac{4q_0L}{k(2n-1)^2\pi^2 I_0((2n-1)\pi\rho_0/L)} \\
&\quad \times I_0\left(\frac{(2n-1)\pi}{L}\rho\right) \cos \frac{(2n-1)\pi}{L} z.
\end{aligned}$$

7. 匀质圆柱，半径为 ρ_0 ，高为 L 。下底保持温度 u_1 ，上底温度分布为 $u_2\rho^2$ ，侧面温度分布为 u_0z 。求解柱体内各点的稳恒温度。

解：根据题意，定解问题是

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = u_0z, \quad u|_{\rho=0} < \infty, \\ u|_{z=0} = u_1, \quad u|_{z=L} = u_2\rho^2. \end{cases} \quad (1)$$

将 u 的定解问题分成两个定解问题，令 $u = v + w$ ，以此代入①，①分解为

$$(1) \quad \begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{\partial^2 v}{\partial z^2} = 0, \\ v|_{\rho=\rho_0} = u_0z, \quad v|_{\rho=0} < \infty, \\ v|_{z=0} = 0, \quad v|_{z=L} = 0. \end{cases} \quad (2)$$

$$(2) \quad \begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial w}{\partial \rho} \right) + \frac{\partial^2 w}{\partial z^2} = 0, \\ w|_{\rho=\rho_0} = 0, \quad w|_{\rho=0} < \infty, \\ w|_{z=0} = u_1, \quad w|_{z=L} = u_2\rho^2. \end{cases} \quad (3)$$

先解 v ，由于上下底面为第一类齐次边值，分离变数后，由 z 方向方程与齐次边值构成本征问题。径向为虚宗量贝塞尔方程，此时，柱内有限解是

$$v(\rho, z) = \sum_{p=1}^{\infty} A_p I_0 \left(\frac{p\pi}{L} \rho \right) \sin \frac{p\pi z}{L}, \quad (4)$$

将④代入②中径向边值，以确定系数，有

$$\sum_{p=1}^{\infty} A_p I_0 \left(\frac{p\pi}{L} \rho_0 \right) \sin \frac{p\pi z}{L} = u_0z,$$

将右端展为正弦级数，比较系数，得

$$\begin{aligned} A_p &= \frac{1}{I_0 \left(\frac{p\pi}{L} \rho_0 \right)} \cdot \frac{2}{L} \int_0^L u_0z \sin \frac{p\pi z}{L} dz \\ &= (-1)^{p+1} \frac{2u_0L}{p\pi I_0 \left(\frac{p\pi}{L} \rho_0 \right)}, \end{aligned}$$

这样

$$v(\rho, z) = \sum_{p=1}^{\infty} \frac{(-1)^{p+1} 2u_0L}{p\pi I_0(p\pi\rho_0/L)} I_0 \left(\frac{p\pi}{L} \rho \right) \sin \frac{p\pi}{L} z.$$

再解 w ，由于侧面为第一类齐次边值，分离变数后，由径向贝塞尔方程与齐次边值构成本征问

题。本证函数与本征值分别为 $J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)$, $\sqrt{\mu_n^{(0)}} = \frac{x_n^{(0)}}{\rho_0}$, $n=1,2,\dots$ 。 $x_n^{(0)}$ 是零阶贝塞尔方程的第 n 个零点。此时, 柱内有限解是

$$w(\rho, z) = \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} z / \rho_0} + B_n e^{-x_n^{(0)} z / \rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right). \quad (5)$$

将⑤代入③中 z 向边值, 以确定系数,

$$\begin{cases} \sum_{n=1}^{\infty} (A_n + B_n) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) = u_1, \\ \sum_{n=1}^{\infty} (A_n e^{x_n^{(0)} L / \rho_0} + B_n e^{-x_n^{(0)} L / \rho_0}) J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) = u_2 \rho^2. \end{cases} \quad (6)$$

将⑥右端按 $J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right)$ 展开, 比较系数, 得

$$\begin{cases} A_n + B_n = \frac{2}{\rho_0^2 [J_1(x_n^{(0)})]^2} \int_0^{\rho_0} u_1 J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \rho d\rho \equiv f_{1n}, \\ A_n e^{x_n^{(0)} L / \rho_0} + B_n e^{-x_n^{(0)} L / \rho_0} \\ = \frac{2}{\rho_0^2 [J_1(x_n^{(0)})]^2} \int_0^{\rho_0} u_2 \rho^2 J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right) \rho d\rho \equiv f_{2n}. \end{cases} \quad (7)$$

具体算出⑦中积分, 得到

$$\begin{cases} f_{1n} = \frac{2u_1}{x_n^{(0)} J_1(x_n^{(0)})}, \\ f_{2n} = \frac{2u_2 \rho_0^2}{x_n^{(0)} J_1(x_n^{(0)})} \left[1 - \frac{4}{(x_n^{(0)})^2} \right]. \end{cases} \quad (8)$$

由⑦解出 A_n 、 B_n

$$\begin{cases} A_n = \frac{f_{2n} - f_{1n} e^{-x_n^{(0)} L / \rho_0}}{2 \text{sh}(x_n^{(0)} L / \rho_0)}, \\ B_n = \frac{f_{1n} e^{x_n^{(0)} L / \rho_0} - f_{2n}}{2 \text{sh}(x_n^{(0)} L / \rho_0)}. \end{cases} \quad (9)$$

将 f_{1n}, f_{2n} 表式⑧代入 A_n, B_n 表式⑨, ⑨再代入⑤, 整理, 即得

$$\begin{aligned} w(\rho, z) = \sum_{n=1}^{\infty} \frac{2}{x_n^{(0)} J_1(x_n^{(0)}) \text{sh} \frac{x_n^{(0)}}{\rho_0} L} & \left\{ u \text{sh} \frac{x_n^{(0)}}{\rho_0} (L - z) \right. \\ & \left. + u_2 \rho_0^2 \left[1 - \frac{4}{(x_n^{(0)})^2} \right] \text{sh} \frac{x_n^{(0)}}{\rho_0} z \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right). \end{aligned}$$

最终

$$\begin{aligned} u(\rho, z) = \sum_{p=1}^{\infty} \frac{(-1)^{p+1} 2u_0 L}{p\pi I_0(p\pi\rho_0/L)} I_0\left(\frac{p\pi}{L}\rho\right) \sin \frac{p\pi}{L} z & + \sum_{n=1}^{\infty} \frac{2}{x_n^{(0)} J_1(x_n^{(0)}) \text{sh} \frac{x_n^{(0)}}{\rho_0} L} \\ & \times \left\{ u_1 \text{sh} \frac{x_n^{(0)}}{\rho_0} (L - z) + u_2 \rho_0^2 \left[1 - \frac{4}{(x_n^{(0)})^2} \right] \text{sh} \frac{x_n^{(0)}}{\rho_0} z \right\} J_0\left(\frac{x_n^{(0)}}{\rho_0}\rho\right). \end{aligned}$$

8. 半径为 ρ_0 、高为 L 的圆柱，上底绝热，下底保持温度 u_0 ，侧面有均匀分布的强度为 q_0 的热流进入，求柱外匀质介质中各点的稳恒温度。

解：根据题意，定解问题是

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial z^2} = 0, \\ u|_{\rho=\rho_0} = \frac{q_0}{k}, \quad u|_{\rho>\rho_0} < \infty, \\ u|_{z=0} = u_0, \quad u_z|_{z=L} = 0. \end{cases} \quad (1)$$

令 $u = u_0 + v$ ，代入①，①转换为 v 的定解问题

$$\begin{cases} \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial v}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 v}{\partial \varphi^2} + \frac{\partial^2 v}{\partial z^2} = 0, \\ v|_{\rho=\rho_0} = \frac{q_0}{k}, \quad v|_{\rho>\rho_0} < \infty, \\ v|_{z=0} = 0, \quad v_z|_{z=L} = 0. \end{cases} \quad (2)$$

对②分离变数后，由 z 向方程及对应齐次边条件构成本征问题

$$\begin{cases} Z'' - \mu Z = 0, \\ Z|_{z=0} = 0, \quad Z_z|_{z=L} = 0. \end{cases}$$

本征函数是 $\sin \frac{(p+1/2)\pi}{L} z$ ，本征值是 $-\mu = \frac{(p+1/2)^2 \pi^2}{L^2}$ ， $p = 0, 1, 2, \dots$ 。径向为虚宗量贝塞尔方程，柱外有限解为虚宗量汉克尔函数，这样，问题②的一般解为

$$v(\rho, z) = \sum_{p=0}^{\infty} A_p K_0 \left(\frac{(p+1/2)\pi}{L} \rho \right) \sin \frac{(p+1/2)\pi}{L} z \quad (3)$$

将③代入②中径向边界条件，以确定系数

$$\sum_{p=0}^{\infty} A_p \frac{(p+1/2)\pi}{L} K'_0 \left(\frac{(p+1/2)\pi}{L} \rho_0 \right) \sin \frac{(p+1/2)\pi}{L} z = \frac{q_0}{k} \quad (4)$$

将④右边按 $\sin \frac{(p+1/2)\pi}{L} z$ 展开，比较系数，得

$$\begin{aligned} A_p &= \frac{2}{L} \cdot \frac{1}{\frac{(p+1/2)\pi}{L} K'_0 \left(\frac{(p+1/2)\pi}{L} \rho_0 \right)} \int_0^L \frac{q_0}{k} \sin \frac{(p+1/2)\pi}{L} z \, dz \\ &= \frac{2q_0 L}{k(p+1/2)^2 \pi^2 K'_0 \left(\frac{(p+1/2)\pi}{L} \rho_0 \right)}, \end{aligned}$$

于是

$$v(\rho, z) = \sum_{p=0}^{\infty} \frac{2q_0 L}{k(p+1/2)^2 \pi^2 K'_0 \left(\frac{(p+1/2)\pi}{L} \rho_0 \right)} K_0 \left(\frac{(p+1/2)\pi}{L} \rho \right) \sin \frac{(p+1/2)\pi}{L} z.$$

结果是

$$\begin{aligned} u(\rho, z) &= u_0 + \sum_{p=0}^{\infty} \frac{2q_0 L}{k(p+1/2)^2 \pi^2 K'_0 \left(\frac{(p+1/2)\pi}{L} \rho_0 \right)} \\ &\quad \times K_0 \left(\frac{(p+1/2)\pi}{L} \rho \right) \sin \frac{(p+1/2)\pi}{L} z. \end{aligned}$$

§11.5 球贝塞尔方程

1. 以 $\psi_l(x)$ 代表 $j_l(x), n_l(x), h_l^{(1)}(x), h_l^{(2)}(x)$ 中任一种, 证明递推关系

$$(1) \quad \psi_{l-1} + \psi_{l+1} = \frac{2l+1}{x} \psi_l,$$

$$(2) \quad l\psi_{l-1} - (l+1)\psi_{l+1} = (2l+1)\psi'_l.$$

证 先证 (1) 式。由教材 (11.1.13) 式, 即

$$Z_{\nu+1}(x) - \frac{2\nu Z_\nu(x)}{x} + Z_{\nu-1}(x) = 0. \quad (1)$$

①式各项同乘以 $\sqrt{\frac{\pi}{2x}}$, 并置 $\nu = l + \frac{1}{2}$, 有

$$\sqrt{\frac{\pi}{2x}} Z_{l+1/2+1}(x) - \sqrt{\frac{\pi}{2x}} \frac{2(l+1/2) Z_{l+1/2}(x)}{x} + \sqrt{\frac{\pi}{2x}} Z_{l+1/2-1}(x) = 0.$$

由 $\psi_l(x) = \sqrt{\frac{\pi}{2x}} Z_{l+\frac{1}{2}}(x)$, 有

$$\psi_{l-1}(x) + \psi_{l+1}(x) = \frac{(2l+1)}{x} \psi_l(x).$$

证毕。

再证 (2) 式

$$\begin{aligned} (2l+1)\psi'_l &= (2l+1) \frac{d}{dx} \left[\sqrt{\frac{\pi}{2x}} Z_{l+1/2}(x) \right] \\ &= (2l+1) \left[\left(-\frac{1}{2} \right) \sqrt{\frac{\pi}{2x^3}} Z_{l+1/2}(x) + \sqrt{\frac{\pi}{2x}} Z'_{l+1/2}(x) \right] \\ &= \sqrt{\frac{\pi}{2x}} \left[-\left(l + \frac{1}{2} \right) \frac{Z_{l+1/2}(x)}{x} + 2 \left(l + \frac{1}{2} \right) Z'_{l+1/2}(x) \right] \end{aligned}$$

由教材 (11.1.12)、(11.1.13) 两式可知

$$\begin{aligned} 2Z'_{l+1/2}(x) &= Z_{l+1/2-1}(x) - Z_{l+1/2+1}(x), \\ \frac{\left(l + \frac{1}{2} \right) Z_{l+1/2}(x)}{x} &= \frac{1}{2} [Z_{l+1/2-1}(x) + Z_{l+1/2+1}(x)], \\ (2l+1)\psi'_l &= \sqrt{\frac{\pi}{2x}} \left[-\frac{1}{2} Z_{l+1+1/2}(x) - \frac{1}{2} Z_{l-1+1/2}(x) \right. \\ &\quad \left. + \left(l + \frac{1}{2} \right) Z_{l-1+1/2}(x) - \left(l + \frac{1}{2} \right) Z_{l+1+1/2}(x) \right] \end{aligned}$$

$$\begin{aligned} &= \sqrt{\frac{\pi}{2x}} [-(l+1)Z_{l+1+1/2}(x) + lZ_{l-1+1/2}(x)] \\ &= -(l+1)\sqrt{\frac{\pi}{2x}} Z_{l+1+1/2}(x) + l\sqrt{\frac{\pi}{2x}} Z_{l-1+1/2}(x) \\ &= l\psi_{l-1}(x) - (l+1)\psi_{l+1}(x) \end{aligned}$$

证毕。

2. 证明

$$(1) [x^3 j_{-1}(x) j_1(x)]' = x^3 j_0(x) [j_{-1}(x) - j_1(x)] = x^2 [x j_0^2(x)]',$$

$$(2) \int x^2 j_0^2(x) dx = \frac{1}{2} x^3 [j_0^2(x) - j_{-1}(x) j_1(x)] + C.$$

证 先证 (1) 式。由教材 (11.5.6) 式知

$$j_{-1}(x) = \frac{\cos x}{x}, \quad j_0(x) = \frac{\sin x}{x}, \quad j_1(x) = \frac{1}{x^2}(\sin x - x \cos x),$$

$$x^3 j_{-1}(x) j_1(x) = x^3 \cdot \frac{\cos x}{x} \cdot \frac{1}{x^2}(\sin x - x \cos x) = \sin x \cos x - x \cos^2 x$$

从而, (1) 式左端为

$$\begin{aligned} [x^3 j_{-1}(x) j_1(x)]' &= (\sin x \cos x - x \cos^2 x)' \\ &= \cos^2 x - \sin^2 x - \cos^2 x + 2x \cos x \sin x \\ &= -\sin^2 x + 2x \cos x \sin x \end{aligned} \quad (1)$$

(1) 式中间项为

$$\begin{aligned} x^3 j_0(x) [j_{-1}(x) - j_1(x)] &= x^3 \frac{\sin x}{x} \left[\frac{\cos x}{x} - \frac{1}{x^2}(\sin x - x \cos x) \right] \\ &= x \sin x \cos x - \sin^2 x + x \sin x \cos x \\ &= 2x \sin x \cos x - \sin^2 x \end{aligned} \quad (2)$$

(1) 式右端为

$$\begin{aligned} x^2 [x j_0^2(x)]' &= x^2 \frac{d}{dx} \left(x \frac{\sin^2 x}{x^2} \right) = x^2 \frac{d}{dx} \left(\frac{\sin^2 x}{x} \right) \\ &= x^2 \frac{d}{dx} \left(\frac{\sin^2 x}{x} \right) = x^2 \frac{2x \sin x \cos x - \sin^2 x}{x^2} \\ &= 2x \sin x \cos x - \sin^2 x \end{aligned} \quad (3)$$

比较①②③式, 可见, (1) 式成立。证毕。

$$\text{再证 (2) 式 } \int x^2 j_0^2(x) dx = \frac{1}{2} x^3 [j_0^2(x) - j_{-1}(x) j_1(x)] + C.$$

由 $j_0(x) = \frac{\sin x}{x}$, 有 $x^2 j_0^2(x) = \sin^2 x$, 于是 (2) 式左边

$$\begin{aligned} \int x^2 j_0^2(x) dx &= \int \sin^2 x dx = \int \frac{1}{2} (1 - \cos 2x) dx \\ &= \frac{1}{2} x - \frac{1}{4} \sin 2x + c = \frac{1}{2} x - \frac{1}{2} \sin x \cos x + c \end{aligned}$$

(2) 式右边

$$\begin{aligned} &\frac{1}{2} x^3 [j_0^2(x) - j_{-1}(x) j_1(x)] + c \\ &= \frac{1}{2} x^3 \left[\frac{\sin^2 x}{x^2} - \frac{\cos x}{x} \frac{1}{x^2} (\sin x - x \cos x) \right] + c \\ &= \frac{1}{2} [x \sin^2 x - \cos x \sin x + x \cos^2 x] + c \\ &= \frac{1}{2} (x - \cos x \sin x) + c \end{aligned}$$

证毕。

3. 证明

$$(1) [x^3 j_{l-1}(x) j_{l+1}(x)]' = x^3 j_l(x) [j_{l-1}(x) - j_{l+1}(x)] = x^2 [x j_l^2(x)]',$$

$$(2) \int x^2 j_l^2(x) dx = \frac{1}{2} x^3 [j_l^2(x) - j_{l-1}(x) j_{l+1}(x)] + C.$$

(提示: 将 (11.5.4) 代入 (11.1.8)、(11.1.9) 得出跟 $j_l(x)$ 有关的求导公式, 利用之)

证 为了证明 (1) (2) 两式, 先根据提示推导跟 $j_l(x)$ 有关的求导公式。由教材 (11.5.4) 式, 即

$$Z_{l+\frac{1}{2}}(x) = \sqrt{\frac{2x}{\pi}} z_l(x) \quad (1)$$

在教材 (11.1.8) (11.1.9) 两式中置 $\nu = l + \frac{1}{2}$, 两式成为

$$\frac{d}{dx} \left[\frac{Z_{l+1/2}(x)}{x^{l+1/2}} \right] = - \frac{Z_{l+1/2+1}(x)}{x^{l+1/2}}, \quad (2)$$

$$\frac{d}{dx} [x^{l+1/2} Z_{l+1/2}(x)] = x^{l+1/2} Z_{l+1/2-1}(x). \quad (3)$$

以①代入②③两式, 得

$$\frac{d}{dx} \left[\frac{\sqrt{\frac{2x}{\pi}} z_l(x)}{x^{l+1/2}} \right] = - \frac{\sqrt{\frac{2x}{\pi}} z_{l+1}(x)}{x^{l+1/2}}, \quad (4)$$

$$\frac{d}{dx} \left[x^{l+1/2} \sqrt{\frac{2x}{\pi}} z_l(x) \right] = x^{l+1/2} \sqrt{\frac{2x}{\pi}} z_{l-1}(x). \quad (5)$$

将④⑤两式左端展开, 此两式又可写成

$$z_l'(x) - \frac{l}{x} z_l(x) = -z_{l+1}(x), \quad (6)$$

$$z_l'(x) + \frac{l+1}{x} z_l(x) = z_{l-1}(x), \quad (7)$$

作⑦+⑥, ⑦-⑥运算, 得

$$2z_l'(x) + \frac{1}{x} z_l(x) = z_{l-1}(x) - z_{l+1}(x), \quad (8)$$

$$\frac{2l+1}{x} z_l(x) = z_{l-1}(x) + z_{l+1}(x), \quad (9)$$

在⑧⑨中将 $z_l \rightarrow j_l$, 并改写成

$$j_l'(x) = \frac{1}{2} j_{l-1}(x) - \frac{1}{2} j_{l+1}(x) - \frac{1}{2x} j_l(x), \quad (10)$$

$$\frac{2l+1}{x} j_l(x) = j_{l-1}(x) + j_{l+1}(x),$$



⑩ 中 $l \rightarrow l-1$ 及 $l \rightarrow l+1$, 分别得到

$$j'_{l-1}(x) = \frac{1}{2} j_{l-2}(x) - \frac{1}{2} j_l(x) - \frac{1}{2x} j_{l-1}(x),$$

$$j'_{l+1}(x) = \frac{1}{2} j_l(x) - \frac{1}{2} j_{l+2}(x) - \frac{1}{2x} j_{l+1}(x),$$

$$j_{l-2}(x) = \frac{2l-1}{x} j_{l-1}(x) - j_l(x),$$

$$j_{l+2}(x) = \frac{2l+3}{x} j_{l+1}(x) - j_l(x),$$

现在来证明第 (1) 式, 其左端

$$[x^3 j_{l-1}(x) j_{l+1}(x)]' = 3x^2 j_{l-1}(x) j_{l+1}(x) + x^3 j'_{l-1}(x) j_{l+1}(x) + x^3 j_{l-1}(x) j'_{l+1}(x)$$

其中 j'_{l-1} 与 j'_{l+1} 以 代入, 中的 j_{l-2} 与 j_{l+2} 以 代入, 经整理即得

$$[x^3 j_{l-1}(x) j_{l+1}(x)]' = x^3 j_l(x) [j_{l-1}(x) - j_{l+1}(x)].$$

再看 (1) 式右端

$$x^2 [x j_l^2(x)]' = x^2 [j_l^2(x) + 2x j_l(x) j'_l(x)]$$

其中

$$2j'_l(x) = j_{l-1}(x) - j_{l+1}(x) - \frac{1}{x} j_l(x),$$

所以

$$\begin{aligned} x^2 [x j_l^2(x)]' &= x^2 \{ j_l^2(x) + x j_l(x) j_{l-1}(x) - x j_l(x) j_{l+1}(x) - j_l^2(x) \} \\ &= x^3 j_l(x) [j_{l-1}(x) - j_{l+1}(x)]. \end{aligned}$$

证毕。

再证明第 (2) 式, $\int x^2 j_l^2(x) dx = \frac{1}{2} x^3 [j_l^2(x) - j_{l-1}(x) j_{l+1}(x)] + C$ 。对等式两边分别求导函数,

左边导函数为 $x^2 j_l^2(x)$ 。右边导函数为

$$\begin{aligned} & \frac{1}{2} \frac{d}{dx} \{ x^3 [j_l^2(x) - j_{l-1}(x) j_{l+1}(x)] + C \} \\ &= \frac{1}{2} \frac{d}{dx} \{ x^3 j_l^2(x) - x^3 j_{l-1}(x) j_{l+1}(x) \} \\ &= \frac{1}{2} \frac{d}{dx} \left\{ \frac{d}{dx} [x^3 j_l^2(x)] - \frac{d}{dx} [x^3 j_{l-1}(x) j_{l+1}(x)] \right\} \end{aligned}$$

式 $\{\dots\}$ 内第二项按 (1) 式为

$$\frac{d}{dx} [x^3 j_{l-1}(x) j_{l+1}(x)] = x^2 \frac{d}{dx} [x j_l^2(x)]$$

以 代入 , 式成为

$$\begin{aligned} & \frac{1}{2} \frac{d}{dx} \{ x^3 [j_l^2(x) - j_{l-1}(x) j_{l+1}(x)] + C \} \\ &= \frac{1}{2} \left\{ \frac{d}{dx} [x^3 j_l^2(x)] - x^2 \frac{d}{dx} [x j_l^2(x)] \right\} \\ &= \frac{1}{2} \{ 3x^2 j_l^2(x) + 2x^3 j_l(x) j'_l(x) - x^2 j_l^2(x) - 2x^3 j_l(x) j'_l(x) \} \\ &= x^2 j_l^2(x). \end{aligned}$$

证毕。

4. 确定球形铀块的临界半径。（“临界”一词参看 § 8.1 习题 8）

解：以球心为坐标系原点建立球坐标系。类似于 § 11.2 第 9 题，定解问题是

$$\begin{cases} u_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \beta u, \\ u|_{r=r_0} = 0, \quad u|_{r=0} < \infty. \end{cases} \quad (1)$$

解法一：令 $u(r, t) = R(r)T(t)$ ，代入定解问题①，分离变数，得

$$T' + k^2 a^2 T = 0, \quad (2)$$

及本征问题

$$\begin{cases} r^2 R'' + 2rR' + \frac{k^2 a^2 + \beta}{a^2} r^2 R = 0 \\ R|_{r=r_0} = 0, \quad R|_{r=0} < \infty, \end{cases}$$

这是零阶球贝塞尔方程在第一类齐次边值条件下的本征值问题。本征函数是

$$R_n(r) = j_0 \left(\frac{\sqrt{(k_n^{(0)})^2 a^2 + \beta}}{a} r \right),$$

本征值 $k_n^{(0)}$ 由

$$j_0 \left(\frac{\sqrt{(k_n^{(0)})^2 a^2 + \beta}}{a} r_0 \right) = \frac{\sin \frac{\sqrt{(k_n^{(0)})^2 a^2 + \beta}}{a} r_0}{\frac{\sqrt{(k_n^{(0)})^2 a^2 + \beta}}{a} r_0} = 0$$

确定，即

$$\frac{\sqrt{(k_n^{(0)})^2 a^2 + \beta}}{a} r_0 = n\pi,$$

亦即

$$(k_n^{(0)})^2 a^2 = \frac{n^2 \pi^2 a^2}{r_0^2} - \beta, \quad (n=1, 2, \dots) \quad (3)$$

方程②的解是 $T(t) = A_n^{(0)} e^{-(k_n^{(0)})^2 a^2 t}$ ，由③式可知，只要有一个 $k_n^{(0)}$ 满足

$$\frac{n^2 \pi^2 a^2}{r_0^2} - \beta < 0,$$

即中子浓度将随时间指数增长，铀块将爆炸。而满足 $\frac{n^2 \pi^2 a^2}{r_0^2} - \beta = 0$ 即 $r_0 = \frac{n\pi a}{\sqrt{\beta}}$ 。取最小值

$n=1$ 时的 r_0 称为临界半径。从而球形铀块的临界半径是 $\frac{\pi a}{\sqrt{\beta}}$ 。

解法二：作函数变换，

$$u(r, \theta, \varphi, t) = v(r, \theta, \varphi, t) e^{\beta t}, \quad (4)$$

以消去方程右端函数项。将式④代入定解问题①，则定解问题①转换成定解问题

$$\begin{cases} v_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) = 0, \\ v|_{r=r_0} = 0, \quad v|_{r=0} < \infty, \end{cases} \quad (5)$$

定解问题⑤在球内分离变数形式的解是

$$v_n(r) = j_0(k_n^{(0)} r) e^{-(k_n^{(0)})^2 a^2 t}.$$

其中 $k_n^{(0)}$ 由 $j_0(k_n^{(0)} r_0) = \frac{\sin k_n^{(0)} r_0}{k_n^{(0)} r_0} = 0$ 确定。有 $k_n^{(0)} = \frac{n\pi}{r_0}$, $n=1, 2, \dots$ 。从而 v 的本征解为

$$v_n(r, t) = j_0 \left(\frac{n\pi}{r_0} r \right) e^{-n^2 \pi^2 a^2 t / r_0^2}$$

而 u 的本征解为

$$u_n(r, t) = A_n j_0 \left(\frac{n\pi}{r_0} r \right) e^{(\beta - n^2 \pi^2 a^2 / r_0^2) t}$$

根据临界半径的定义，有球形铀块临界半径为 $\pi a / \sqrt{\beta}$ 。（ $n=1$ ）

5. 均质球，半径为 r_0 ，初始温度分布为 $f(r)$ 。把球面温度保持为零度而使它冷却。求解球内各处温度变化情况。

解：根据题意，球内各处温度与角度 θ, φ 无关，定解问题是

$$\begin{cases} u_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = 0, \\ u|_{r=r_0} = 0, \quad u|_{r=0} < \infty, \\ u|_{t=0} = f(r), \end{cases} \quad (1)$$

令 $u(r, t) = R(r)T(t)$ ，代入定解问题 (1)，分离变数，得

$$T' + k^2 a^2 T = 0, \quad (2)$$

及本征问题

$$\begin{cases} r^2 R'' + 2rR' + k^2 r^2 R = 0, \\ R|_{r=r_0} = 0, \quad R|_{r=0} < \infty, \end{cases}$$

这是零阶球贝塞尔方程在第一类齐次边值条件下的本征值问题。本征函数是 $R_n(r) = j_0(k_n^{(0)}r)$ ，

本征值 $k_n^{(0)}$ 由 $j_0(k_n^{(0)}r_0) = \frac{\sin k_n^{(0)}r_0}{k_n^{(0)}r_0} = 0$ 确定，即 $k_n^{(0)}r_0 = n\pi$ ， $k_n^{(0)} = \frac{n\pi}{r_0}$ ， $n = 1, 2, \dots$ ，方程 (2) 的

解是

$$T(t) = Be^{-k_n^{(0)2}a^2t} = Be^{-\frac{n^2\pi^2a^2}{r_0^2}t}。$$

从而 (1) 的一般解为

$$u(r, t) = \sum_{n=1}^{\infty} A_n \frac{\sin(n\pi r / r_0)}{n\pi r / r_0} e^{-(n\pi/r_0)^2 a^2 t}。 \quad (3)$$

将 (3) 代入 (1) 中初始条件以确定系数

$$\sum_{n=1}^{\infty} A_n \frac{\sin(n\pi r / r_0)}{n\pi r / r_0} = f(r)。 \quad (4)$$

将 (4) 右边的 $f(r)$ 按球贝塞尔函数展开，比较两边系数，

$$A_n = \frac{1}{N_n^2} \int_0^{r_0} f(r) \frac{\sin(n\pi r / r_0)}{n\pi r / r_0} r^2 dr。$$

其中

$$N_n^2 = \int_0^{r_0} [j_0(k_n r)]^2 r^2 dr = \int_0^{r_0} \left[\frac{\sin n\pi r / r_0}{n\pi r / r_0} \right]^2 r^2 dr = \frac{r_0^3}{2n^2\pi^2}。$$

于是

$$A_n = \frac{2n^2\pi^2}{r_0^3} \cdot \frac{r_0}{n\pi} \int_0^{r_0} f(r) r \sin \frac{n\pi r}{r_0} dr = \frac{2n\pi}{r_0^2} \int_0^{r_0} r f(r) \sin \frac{n\pi r}{r_0} dr。$$

这样，本例的答案是

$$\begin{aligned} u(r, t) &= \sum_{n=1}^{\infty} \frac{2n\pi}{r_0^2} \int_0^{r_0} r f(r) \sin \frac{n\pi r}{r_0} dr j_0\left(\frac{n\pi}{r_0}r\right) e^{-\frac{n^2\pi^2a^2}{r_0^2}t} \\ &= \frac{2}{r_0} \cdot \frac{1}{r} \sum_{n=1}^{\infty} \int_0^{r_0} r f(r) \sin \frac{n\pi r}{r_0} dr \cdot \sin \frac{n\pi r}{r_0} \cdot e^{-\frac{n^2\pi^2a^2}{r_0^2}t}。 \end{aligned}$$

6. 均质球，半径为 r_0 ，初始温度分布为 $f(r)\cos\theta$ 。把球面温度保持为零度而使它冷却，求解球内各处温度变化情况。

解：类似第 5 题，定解问题是

$$\begin{cases} u_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial u}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) \right] = 0, \\ u|_{r=r_0} = 0, \quad u|_{r=0} < \infty, \\ u|_{t=0} = f(r)\cos\theta, \end{cases} \tag{1}$$

由于初始条件中含有因子 $\cos\theta$ ，即 $P_1(\cos\theta)$ ，于是，径向由一阶球贝塞尔方程与对应齐次边值构成本证问题，本征函数为 $j_1(k_n r)$ ，本征值 k_n ($n=1,2,\dots$) 由

$$j_1(k_n r_0) = \frac{\sin k_n r_0 - k_n r_0 \cos k_n r_0}{(k_n r_0)^2} = 0 \text{ , 即 } \operatorname{tg} k_n r_0 = k_n r_0 \text{ 确定。于是，问题①的一般解为}$$

$$u(r,t) = \sum_{n=1}^{\infty} A_n j_1(k_n r) P_1(\cos \theta) e^{-a^2 k_n^2 t} \tag{2}$$

将②代入①中初始条件确定系数

$$\sum_{n=1}^{\infty} A_n j_1(k_n r) P_1(\cos \theta) = f(r) P_1(\cos \theta) \text{ 。}$$

系数

$$A_n = \frac{1}{N_n^2} \int_0^{r_0} f(r) j_1(k_n r) r^2 \mathrm{d}r$$

其中

$$\begin{aligned} N_n^2 &= \int_0^{r_0} [j_1(k_n r)]^2 r^2 \mathrm{d}r = \int_0^{r_0} \left[\sqrt{\frac{\pi}{2k_n r}} J_{3/2}(k_n r) \right]^2 r^2 \mathrm{d}r \\ &= \frac{\pi}{2k_n} \int_0^{r_0} [J_{3/2}(k_n r)]^2 r \mathrm{d}r \\ &= \frac{\pi r_0^2}{4k_n} [J'_{3/2}(k_n r_0)]^2 \\ &= \frac{\pi r_0^2}{4k_n} \left[J_{1/2}(k_n r_0) - \frac{3}{2} \frac{J_{3/2}(k_n r_0)}{k_n r_0} \right]^2 \\ &= \frac{\pi r_0^2}{4k_n} [J_{1/2}(k_n r_0)]^2 \quad (\text{其中 } J_{3/2}(k_n r_0) : j_1(k_n r_0) = 0) \\ &= \frac{r_0^3}{2} \left[\sqrt{\frac{\pi}{2k_n r_0}} J_{1/2}(k_n r_0) \right]^2 \\ &= \frac{r_0^3}{2} [j_0(k_n r_0)]^2 \text{ 。} \end{aligned}$$

结果

$$u(r,t) = \frac{2}{r_0^3} \sum_{n=1}^{\infty} \frac{j_1(k_n r)}{[j_0(k_n r_0)]^2} P_1(\cos \theta) e^{-a^2 k_n^2 t} \int_0^{r_0} j_1(k_n r) f(r) r^2 \mathrm{d}r \text{ 。}$$

7. 半径为 $2r_0$ 的均质球, 初始温度 = $\begin{cases} u_0, & (0 < r < r_0), \\ 0, & (r_0 < r < 2r_0). \end{cases}$ 把球面保持为零度而使它冷却,

求解球内温度变化情况。

解: 根据题意, 球内各处温度与角度 θ, φ 无关, 定解问题是

$$\begin{cases} u_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = 0, \\ u|_{r=2r_0} = 0, \quad u|_{r=0} < \infty, \\ u|_{t=0} = \begin{cases} u_0 & (0 < r < r_0) \\ 0 & (r_0 < r < 2r_0) \end{cases}. \end{cases} \quad (1)$$

令 $u(r, t) = R(r)T(t)$, 代入定解问题①, 分离变数, 得

$$T' + k^2 a^2 T = 0, \quad (2)$$

及本征问题

$$\begin{cases} r^2 R'' + 2rR' + k^2 r^2 R = 0 \\ R|_{r=r_0} = 0, \quad R|_{r=0} < \infty, \end{cases}$$

这是零阶球贝塞尔方程在第一类齐次边值条件下的本征值问题。本征函数是 $R_n(r) = j_0(k_n^{(0)}r)$,

本征值 $k_n^{(0)}$ 由 $j_0(2k_n^{(0)}r_0) = \frac{\sin 2k_n^{(0)}r_0}{2k_n^{(0)}r_0} = 0$ 确定, 即

$$2k_n^{(0)}r_0 = n\pi, \quad k_n^{(0)} = \frac{n\pi}{2r_0}, \quad n = 1, 2, \dots,$$

方程②的解是

$$T(t) = Be^{-k_n^{(0)2}a^2t} = Be^{-\frac{n^2\pi^2a^2}{(2r_0)^2}t}.$$

这样, u 的一般解是

$$u(r, t) = \sum_{n=1}^{\infty} A_n j_0\left(\frac{n\pi r}{2r_0}\right) e^{-(n\pi/2r_0)^2 a^2 t}. \quad (3)$$

将③代入①中初始条件以确定系数

$$\sum_{n=1}^{\infty} A_n j_0\left(\frac{n\pi r}{2r_0}\right) = \begin{cases} u_0 & (0 < r < r_0) \\ 0 & (r_0 < r < 2r_0) \end{cases}$$

将上式右边的 $f(r)$ 按球贝塞尔函数展开, 比较两边系数, 有

$$A_n = \frac{1}{N_n^2} \int_0^{r_0} u_0 \frac{\sin k_n r}{k_n r} r^2 dr$$

其中

$$\begin{aligned}
N_n^2 &= \int_0^{2r_0} [j_0(k_n r)]^2 r^2 dr = \int_0^{2r_0} \left[\frac{\sin k_n r}{k_n r} \right]^2 r^2 dr \\
&= \frac{1}{k_n^2} \int_0^{2r_0} \sin^2 k_n r dr \\
&= \frac{1}{k_n^2} \cdot \frac{1}{k_n} \int_0^{2r_0} \sin^2 k_n r d(k_n r) \\
&= \frac{1}{k_n^3} \left[\frac{1}{2} k_n r - \frac{1}{2} \sin k_n r \cos k_n r \right]_0^{2r_0} \\
&= \frac{r_0}{k_n^2}
\end{aligned}$$

$$\begin{aligned}
\int_0^{r_0} u_0 \frac{\sin k_n r}{k_n r} r^2 dr &= \frac{u_0}{k_n} \int_0^{r_0} r \sin k_n r dr \\
&= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin k_n r - \frac{1}{k_n} \cdot r \cdot \cos k_n r \right]_0^{r_0} \\
&= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin k_n r_0 - \frac{1}{k_n} \cdot r_0 \cdot \cos k_n r_0 \right] \\
&= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin k_n r_0 - \frac{1}{k_n} \cdot r_0 \cdot \cos k_n r_0 \right] \\
&= \frac{u_0}{k_n} \left[\frac{1}{k_n^2} \sin \left(\frac{n\pi}{2} \right) - \frac{r_0}{k_n} \cos \left(\frac{n\pi}{2} \right) \right] \\
&= \frac{u_0}{k_n^2} \left[\frac{1}{k_n} \sin \left(\frac{n\pi}{2} \right) - r_0 \cos \left(\frac{n\pi}{2} \right) \right]
\end{aligned}$$

从而

$$\begin{aligned}
A_n &= \frac{u_0}{k_n^2} \left[\frac{1}{k_n} \sin \left(\frac{n\pi}{2} \right) - r_0 \cos \left(\frac{n\pi}{2} \right) \right] / \frac{r_0}{k_n^2} \\
&= \frac{u_0}{r_0} \left[\frac{1}{k_n} \sin \left(\frac{n\pi}{2} \right) - r_0 \cos \left(\frac{n\pi}{2} \right) \right] \\
&= \frac{u_0}{r_0 k_n} \sin \left(\frac{n\pi}{2} \right) - u_0 \cos \left(\frac{n\pi}{2} \right)
\end{aligned}$$

于是，解为

$$\begin{aligned}
u(r,t) &= \sum_{n=1}^{\infty} \left[\frac{u_0}{r_0 k_n} \sin\left(\frac{n\pi}{2}\right) - u_0 \cos\left(\frac{n\pi}{2}\right) \right] \frac{\sin k_n r}{k_n r} e^{-(n\pi/2r_0)^2 a^2 t} \\
&= \sum_{n=1}^{\infty} \left[\frac{4r_0 u_0}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) - \frac{2r_0 u_0}{n\pi} \cos\left(\frac{n\pi}{2}\right) \right] \frac{\sin k_n r}{r} e^{-(n\pi/2r_0)^2 a^2 t} \\
&= \sum_{n=1}^{\infty} \left[\frac{4r_0 u_0}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right) - \frac{2r_0 u_0}{n\pi} \cos\left(\frac{n\pi}{2}\right) \right] \frac{\sin k_n r}{r} e^{-(n\pi/2r_0)^2 a^2 t} \\
&= \sum_{n=1}^{\infty} A_n \frac{\sin k_n r}{r} e^{-(n\pi/2r_0)^2 a^2 t} \\
&= \sum_{n=1}^{\infty} A_n \frac{1}{r} \sin \frac{n\pi r}{2r_0} e^{-(n\pi/2r_0)^2 a^2 t}. \\
u(r,t) &= \sum_{n=1}^{\infty} A_n \frac{1}{r} \sin \frac{n\pi r}{2r_0} e^{-(n^2 \pi^2 a^2 / 4r_0^2) t},
\end{aligned}$$

其中

$$\begin{aligned}
A_{2k} &= (-1)^{k+1} \frac{u_0 r_0}{k\pi}, \quad A_{2k+1} = (-1)^k \frac{4u_0 r_0}{(2k+1)^2 \pi^2}, \\
\text{Q} \quad \sin \frac{n\pi}{2} &= \begin{cases} 0, & (n=2k), \\ (-1)^k, & (n=2k+1), \end{cases} \\
\cos \frac{n\pi}{2} &= \begin{cases} 0, & (n=2k+1), \\ (-1)^k, & (n=2k). \end{cases}
\end{aligned}$$

8. 均质球, 半径为 r_0 , 初始温度为 U_0 。放在温度为 u_0 的空气中自由冷却 (按照牛顿冷却定律跟空气交换热量), 求解球内各处温度变化情况。

解: 根据题意, 球内各处温度与角度 θ, φ 无关, 定解问题是

$$\begin{cases} u_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = 0, \\ (u + Hu_r)|_{r=r_0} = u_0, \quad H = \frac{k}{h}, \quad u|_{r=0} < \infty, \\ u|_{t=0} = U_0. \end{cases} \quad (1)$$

先齐次化边界条件。令 $u(r, t) = u_0 + v(r, t)$, 代入定解问题①, ①转换成 v 的定解问题

$$\begin{cases} v_t - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) = 0, \\ (v + Hv_r)|_{r=r_0} = 0, \quad H = \frac{k}{h}, \quad v|_{r=0} < \infty, \\ v|_{t=0} = U_0 - u_0. \end{cases} \quad (2)$$

令 $v(r, t) = R(r)T(t)$, 代入定解问题②, 分离变数, 得

$$T' + k^2 a^2 T = 0, \quad (3)$$

及本征问题

$$\begin{cases} r^2 R'' + 2rR' + k^2 r^2 R = 0 \\ (R + HR_r)|_{r=r_0} = 0, \quad H = \frac{k}{h}, \quad R|_{r=0} < \infty, \end{cases}$$

这是零阶球贝塞尔方程在第一类齐次边值条件下的本征值问题。本征函数是 $R_n(r) = j_0(k_n r)$,

本征值 k_n 由 $\left[\frac{\sin k_n r}{k_n r} + H \frac{d}{dr} \frac{\sin k_n r}{k_n r} \right]_{r=r_0} = 0$ 确定, 即

$$k_n r_0 \sin k_n r_0 + H k_n^2 r_0 \cos k_n r_0 - H k_n \sin k_n r_0 = 0, \quad (4)$$

由④得到

$$\operatorname{tg} k_n r_0 = \frac{H k_n r_0}{H - r_0}, \quad n = 1, 2, \dots$$

其中 $k_n = x_n / r_0$, 而 x_n 是方程 $x + \eta \operatorname{tg} x = 0$ 的第 n 个根, $\eta = (H - r_0) / H$ 。 T 的方程③的解是

$$T(t) = B e^{-k_n^2 a^2 t}$$

从而 v 的一般解是

$$v(r, t) = \sum_{n=1}^{\infty} A_n \frac{\sin k_n r}{k_n r} e^{-k_n^2 a^2 t} \quad (5)$$

将⑤代入②中初始条件以确定系数

$$\sum_{n=1}^{\infty} A_n \frac{\sin k_n r}{k_n r} = U_0 - u_0$$

将上式右边的 $U_0 - u_0$ 按球贝塞尔函数展开, 比较两边系数,

$$A_n = \frac{U_0 - u_0}{N_n^2} \int_0^{r_0} \frac{\sin k_n r}{k_n r} r^2 dr.$$

其中积分

$$(1) \quad N_n^2 = \int_0^{r_0} [j_0(k_n r)]^2 r^2 dr = \int_0^{r_0} \left[\frac{\sin k_n r}{k_n r} \right]^2 r^2 dr$$

$$= \frac{1}{k_n^2} \int_0^{r_0} \sin^2 k_n r dr$$

$$= \frac{1}{k_n^2} \cdot \frac{1}{k_n} \int_0^{r_0} \sin^2 k_n r d(k_n r)$$

$$= \frac{1}{k_n^3} \left[\frac{1}{2} k_n r - \frac{1}{2} \sin k_n r \cos k_n r \right]_0^{r_0}$$

$$= \frac{1}{2k_n^3} [k_n r_0 - \sin k_n r_0 \cos k_n r_0]$$

$$(2) \quad (U_0 - u_0) \int_0^{r_0} \frac{\sin k_n r}{k_n r} r^2 dr = \frac{(U_0 - u_0)}{k_n} \int_0^{r_0} r \sin k_n r dr$$

$$= \frac{(U_0 - u_0)}{k_n} \left[\frac{1}{k_n^2} \sin k_n r - \frac{1}{k_n} \cdot r \cdot \cos k_n r \right]_0^{r_0}$$

$$= \frac{(U_0 - u_0)}{k_0} \left[\frac{1}{k_n^2} \sin k_n r_0 - \frac{1}{k_n} \cdot r_0 \cdot \cos k_n r_0 \right]$$

$$= \frac{(U_0 - u_0)}{k_0^2} \left[\frac{1}{k_n} \sin k_n r_0 - r_0 \cdot \cos k_n r_0 \right]$$

从而

$$A_n = \frac{U_0 - u_0}{N_n^2} \int_0^{r_0} \frac{\sin k_n r}{k_n r} r^2 dr$$

其中

$$\begin{aligned}
\int_0^{r_0} \frac{\sin k_n r}{k_n r} r^2 dr &= \frac{1}{k_n} \int_0^{r_0} r \sin k_n r dr \\
&= \frac{1}{k_n} \left[\frac{1}{k_n^2} \sin k_n r - \frac{1}{k_n} \cdot r \cdot \cos k_n r \right]_0^{r_0} \\
&= \frac{1}{k_n} \left[\frac{1}{k_n^2} \sin k_n r_0 - \frac{1}{k_n} \cdot r_0 \cdot \cos k_n r_0 \right] \\
&= \frac{1}{k_n^2} \left[\sin k_n r_0 - r_0 \cdot \cos k_n r_0 \right]
\end{aligned}$$

这样

$$\begin{aligned}
A_n &= \frac{(U_0 - u_0)}{k_n^2} \left[\frac{1}{k_n} \sin k_n r_0 - r_0 \cos k_n r_0 \right] \bigg/ \frac{1}{2k_n^3} [k_n r_0 - \sin k_n r_0 \cos k_n r_0] \\
&= 2(U_0 - u_0) \frac{\sin k_n r_0 - k_n r_0 \cos k_n r_0}{k_n r_0 - \sin k_n r_0 \cos k_n r_0} \\
&= \frac{2r_0(U_0 - u_0)}{H} \frac{\sin k_n r_0}{k_n r_0 - (\sin 2k_n r_0)/2}
\end{aligned}$$

解为

$$u(r, t) = u_0 + \sum_{n=1}^{\infty} \frac{2r_0(U_0 - u_0)}{H} \cdot \frac{\sin k_n r_0}{k_n r_0 - (\sin 2k_n r_0)/2} \cdot \frac{\sin k_n r}{k_n r} e^{-k_n^2 a^2 t},$$

其中 $k_n = x_n/r_0$, 而 x_n 是方程 $x + \eta \tan x = 0$ 的第 n 个根, $\eta = (r_0 - H)/H$ 。

9. 半径为 r_0 的球面径向速度分布为 $v = v_0 \frac{1}{4} (3 \cos 2\theta + 1) \cos \omega t$ 。试求解这个球在空气中辐射出去的声场中的速度势，设 $r_0 = \lambda$ （声波波长）。本题径向速度对空间中的方向的依赖性由因子 $\frac{1}{4} (3 \cos 2\theta + 1)$ 即 $P_2(\cos \theta)$ 描写，因而是轴对称四极声源。

解：本例跟 § 11.5 例 2 类似，也是没有初始条件的问题。用球坐标系，极点取在球心。定解问题是

$$\begin{cases} u_{tt} - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial u}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) \right] = 0, \\ u_r|_{r=r_0} = v_0 \frac{1}{4} (3 \cos 2\theta + 1) e^{-i\omega t} = v_0 P_2(\cos \theta) e^{-i\omega t}, \end{cases} \quad (1)$$

边界条件里的 $\cos \omega t$ 即 $\text{Re}(e^{-i\omega t})$ 。写成了 $e^{-i\omega t}$ ，在结果中取其实部即为问题的解。设波动解为 $v(r, \theta) = A h_2^{(1)}(kr) P_2(\cos \theta)$ 。

代入定解问题 (1)，有

$$\begin{cases} \nabla^2 v + k^2 v = 0, \quad (k = \frac{\omega}{a}, \quad r > r_0) \\ \left. \frac{\partial v(r)}{\partial r} \right|_{r=r_0} = v_0 P_2(\cos \theta) \end{cases} \quad (2)$$

对 (2) 进一步分离变数，并考虑边界条件，得 v 的解

$$\begin{aligned} v(r, \theta) &= A h_2^{(1)}(kr) P_2(\cos \theta) \\ &= A \left(-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right) e^{ikr} P_2(\cos \theta), \end{aligned} \quad (3)$$

将 (3) 代入 (2) 中边界条件以确定系数 A

$$A \left(\frac{9i}{(kr_0)^4} + \frac{9}{(kr_0)^3} - \frac{4i}{(kr_0)^2} - \frac{1}{kr_0} \right) k e^{ikr_0} = v_0$$

由于 $r_0 = \lambda$ ，即 $kr_0 = 2\pi$ ， kr_0 很小，同时， $e^{ikr_0} \approx 1$ ，于是，上式可以第一项近似表示，

$$A \frac{9i}{(kr_0)^4} k = v_0。$$

从而

$$A = -i \frac{v_0 k^3 r_0^4}{9},$$

于是

$$u(r, \theta, t) = -i \frac{v_0 k^3 r_0^4}{9} h_2^{(1)}(kr) P_2(\cos \theta) e^{-i\omega t}。$$

取实部

$$u(r, \theta, t) = \text{Re} \left\{ -i \frac{v_0 k^3 r_0^4}{9} h_2^{(1)}(kr) P_2(\cos \theta) e^{-i\omega t} \right\}。$$

在远场区，($r \gg 0$)

$$u(r, \theta, t) = -i \frac{v_0 k^3 r_0^4}{9} \left(-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right) e^{ikr} P_2(\cos \theta) e^{-i\omega t}。$$

由于 $r \gg 0$ ，括号内可仅取第三项作为近似

$$u(r, \theta, t) = \frac{v_0 k^2 r_0^4}{9r} P_2(\cos \theta) e^{ik(r-at)}。$$

取实部

$$u(r, \theta, t) = \frac{v_0 k^2 r_0^4}{9r} P_2(\cos \theta) \cos k(r-at)。$$

10. 半径为 r_0 的球面径向速度分布为 $v = v_0 \frac{3}{2}(1 - \cos 2\theta) \sin 2\varphi \cos \omega t$ 。试求解这个球在空气中辐射出去的声场中的速度势, 设 $r_0 = \lambda$ (声波波长)。本题是非轴对称的四极声源。

解: 本例跟 § 11.5 例 2 类似, 也是没有初始条件的问题。用球坐标系, 极点取在球心。定解问题是

$$\begin{cases} u_{tt} - a^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 \frac{\partial u}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) \right] = 0, \\ u_r|_{r=r_0} = v_0 \frac{3}{2}(1 - \cos 2\theta) \sin 2\varphi e^{-i\omega t} = v_0 P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t}. \end{cases}$$

① 边界条件里的 $\cos \omega t$ 即 $\text{Re}(e^{-i\omega t})$, 写成了 $e^{-i\omega t}$, 在结果中取其实部即为问题的解。设波动解为

$$u(r, \theta, \varphi, t) = v(r, \theta, \varphi) e^{-i\omega t}. \quad (2)$$

将②代入定解问题①, 有

$$\begin{cases} \nabla^2 v + k^2 v = 0, & (k = \frac{\omega}{a}, r > r_0) \\ \frac{\partial v}{\partial r} \Big|_{r=r_0} = v_0 P_2^2(\cos \theta) \sin 2\varphi \end{cases} \quad (3)$$

对③进一步分离变数, 并与边界条件比较, v 的解为

$$\begin{aligned} v(r, \theta) &= A h_2^{(1)}(kr) P_2^2(\cos \theta) \sin 2\varphi \\ &= A \left(-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right) e^{ikr} P_2^2(\cos \theta) \sin 2\varphi \end{aligned} \quad (4)$$

将④代入③中边界条件以确定系数 A

$$A \left(\frac{9i}{(kr_0)^4} + \frac{9}{(kr_0)^3} - \frac{4i}{(kr_0)^2} - \frac{1}{kr_0} \right) k e^{ikr_0} = v_0 \quad (5)$$

由于 $r_0 = \lambda$, 即 $kr_0 = 2\pi$, kr_0 很小, 同时, $e^{ikr_0} \approx 1$, 于是, ⑤式可以第一项近似表示

$$A \frac{9i}{(kr_0)^4} k = v_0 \Rightarrow A = -i \frac{v_0 k^3 r_0^4}{9},$$

从而

$$u(r, \theta, \varphi, t) = -i \frac{v_0 k^3 r_0^4}{9} h_2^{(1)}(kr) P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t}.$$

取实部

$$u(r, \theta, \varphi, t) = \text{Re} \left\{ -i \frac{v_0 k^3 r_0^4}{9} h_2^{(1)}(kr) P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t} \right\}.$$

在远场区, ($r \gg 0$)

$$u(r, \theta, \varphi, t) = -i \frac{v_0 k^3 r_0^4}{9} \left(-\frac{3i}{(kr)^3} - \frac{3}{(kr)^2} + \frac{i}{kr} \right) e^{ikr} P_2^2(\cos \theta) \sin 2\varphi e^{-i\omega t}.$$

由于 $r \gg 0$, 括号内可仅取第三项作为近似

$$u(r, \theta, \varphi, t) = \frac{v_0 k^2 r_0^4}{9r} P_2^2(\cos \theta) \sin 2\varphi e^{ik(r-at)}.$$

取实部

$$u(r, \theta, \varphi, t) = \frac{v_0 k^2 r_0^4}{9r} P_2^2(\cos \theta) \sin 2\varphi \cos k(r-at).$$